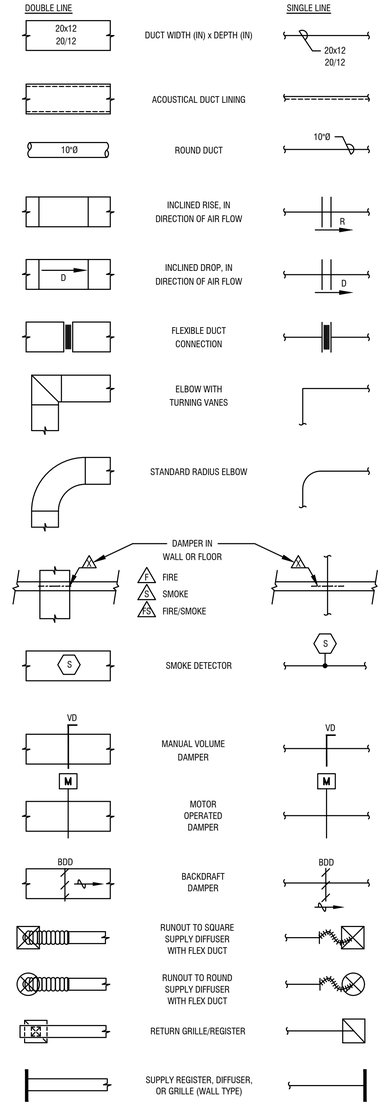
MECHANICAL ABBREVIATIONS

AC	ALTERNATING CURRENT
ACC	AIR COOLED CONDENSER
ACU	AIR CONDITIONING UNIT
AD	ACCESS DOOR
ADJ	ADJUSTABLE
AF	AIR FOIL
AFF	ABOVE FINISHED FLOOR
AHU	AIR HANDLING UNIT
AP	ACCESS PANEL
ARCH	ARCHITECTURE
ATC	AUTOMATIC TEMPERATURE CONTROL
ATM	ATMOSPHERE
AUX	AUXILIARY
AVG	AVERAGE
AWG	AMERICAN WIRE GAUGE
B	BOILER
BDD	BACKDRAFT DAMPER
BHP	BRAKE HORSEPOWER
BOD	BOTTOM OF DUCT
BOS	BOTTOM OF STEEL
BTU	BRITISH THERMAL UNIT
BTUH	BTU PER HOUR
CAV	CONSTANT AIR VOLUME
CC	COOLING COIL
CFM/ ∮	CUBIC FEET PER MINUTE ∮
CH	CHILLER
C&I	CONTROLS & INSTRUMENTATION
CLG	CEILING
CMU	CONCRETE MASONRY UNIT
C0	CARBON MONOXIDE
C02	CARBON DIOXIDE
COND	CONDENSATE
CONN	CONNECTION
CONT.	CONTINUATION
CORR	CORRIDOR
CT	COOLING TOWER
CU	CONDENSING UNIT
CUH	CABINET UNIT HEATER
CV	CONTROL VALVE
CU FT	CUBIC FEET
CVS	CONTROL VALVE STATION
D	DIFFUSER OR REGISTER
dB	DECIBEL, RE 10 WATT
DBT	DRY BULB TEMPERATURE (°F)
DDC	DIRECT DIGITAL CONTROL
DEG	DEGREE FAHRENHEIT (°F)
DIA	DIAMETER
DIM	DIMENSION
DN	DOWN
DP	DEW POINT TEMPERATURE (°F)
DX	DIRECT EXPANSION
DWG	DRAWING
E	EXHAUST
EA	EXHAUST AIR
EAT	ENTERING AIR TEMPERATURE (°F)
EC	ELECTRICAL CONTRACTOR
EER	ENERGY EFFICIENCY RATIO
EF	EXHAUST FAN
EFF	EFFICIENCY
ELEV	ELEVATION
ESP	EXTERNAL STATIC PRESSURE
ET	EXPANSION TANK
EWT	ENTERING WATER TEMPERATURE (°F)
F	FILTER
FA	FACE AREA
FC	FORWARD CURVED
FCU	FAN COIL UNIT
FD	FLOOR DRAIN
FLR	FLOOR
FP	FIRE PROTECTION
FPM	FEET PER MINUTE
FR	FINNED RADIATION
FT	FEET
FVEL	FACE VELOCITY
F&T	FLOAT AND THERMOSTATIC
GA	GAUGE OR GAGE
GAL	GALLON
GALV	GALVANIZED
GC	GENERAL CONTRACTOR
GH	GRAVITY HOOD
GPM	GALLONS PER MINUTE
HC	HEATING COIL
HOA	HAND-OFF-AUTOMATIC
HP	HORSEPOWER
HR	HOUR
HVU	HEATING & VENTILATING UNIT
HWST	HOT WATER SUPPLY TEMPERATURE
HX	HEAT EXCHANGER
IB	INVERTED BUCKET TRAP
IFB	INTEGRAL FACE & BYPASS
IN	INCH
IPS	IRON PIPE SIZE
KW	KILOWATTS
KWH	KILOWATT HOUR
LAT	LEAVING AIR TEMPERATURE (°F)
LBS/#	POUNDS
LD	LINEAR DIFFUSER
LF	LINEAR FEET
LWT	LEAVING WATER TEMPERATURE (°F)
MOD	MOTOR OPERATED DAMPER
MAU	MAKEUP AIR UNIT
MAX	MAXIMUM
MB	MIXING BOX
MBH	1000 BTUH
MC	MECHANICAL CONTRACTOR
MCC	MOTOR CONTROL CENTER
MFR	MANUFACTURER
MIN	MINIMUM
MISC	MISCELLANEOUS
MMBH	1,000,000 BTUH
NA	NOT APPLICABLE
NC	NOISE CRITERIA, dB RE 20 uPa
NC	NORMALLY CLOSED
NIC	NOT IN CONTRACT
NO	NORMALLY OPEN
NTS	NOT TO SCALE
OA	OUTSIDE AIR
OAT	OUTSIDE AIR TEMPERATURE
OAD	OUTSIDE AIR DAMPER
OBD	OPPOSED BLADE DAMPER
OZ	OUNCE
P	PUMP
PC	PLUMBING CONTRACTOR
PD	PRESSURE DROP
PH	PHASE
PHC	PREHEAT COIL
PROP	PROPELLER
PSI	POUNDS PER SQUARE INCH
PSIG	PSI, GAUGE
P+F	PITCH AND FLOW
QTY	QUANTITY
R	REGISTER
RA	RETURN AIR
RC	ROOM CRITERIA, dB RE 20 uPa
RF	RETURN/RELIEF AIR FAN
RH	RELATIVE HUMIDITY
RHC	REHEAT COIL
RPM	REVOLUTIONS PER MINUTE
S	SMOKE DETECTOR
SA	SUPPLY AIR
SEER	SEASONAL ENERGY EFFICIENCY RATIO
SENS	SENSIBLE
SF	SQUARE FEET
SP	STATIC PRESSURE (IN. WG.)
SPEC	SPECIFICATION
SQ	SQUARE
SS	STAINLESS STEEL
SV	SOLENOID VALVE
T	THERMOSTAT
TEMP	TEMPERATURE (°F)
TOD	TOP OF DUCT
TOP	TOP OF PIPE
TOS	TOP OF STEEL
TSP	TOTAL STATIC PRESSURE
TSTAT	THERMOSTAT
TxD	TRANSFER DUCT
TYP	TYPICAL
UC	UNDERCUT
UH	UNIT HEATER
UL	UNDERWRITERS LABORATORY
UV	UNIT VENTILATOR
UVL	ULTRA VIOLET LIGHT
V	VALVE, VOLT
VA	VOLT-AMPERE
VD	VOLUME DAMPER
VEL	VELOCITY
VFD	VARIABLE FREQUENCY DRIVE
VFV	VAV, WITH FAN
VOL	VOLUME
W	WATT
WB	WET BULB TEMPERATURE (°F)
WC	WATER COLUMN
WG	WATER GAUGE
WMS	WIRE MESH SCREEN

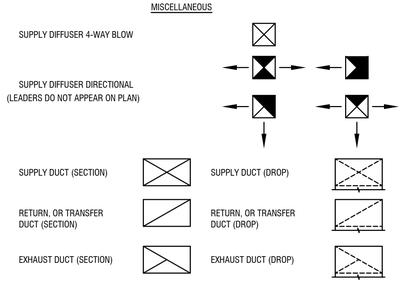
ALL INSTALLATIONS AND MATERIALS SHALL MEET THE FOLLOWING:

1. NEW YORK STATE BUILDING CODE; 2020
2. NEW YORK STATE FIRE CODE; 2020
3. INTERNATIONAL ENERGY CONSERVATION CODE; 2020 WITH NEW YORK STATE SUPPLEMENTS
4. NEW YORK STATE MECHANICAL CODE; 2020
5. NEW YORK STATE FUEL GAS CODE; 2020
6. ALL FEDERAL, STATE AND LOCAL ORDINANCES

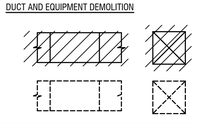
DUCTWORK SYMBOLS



MISCELLANEOUS



DUCT AND EQUIPMENT DEMOLITION



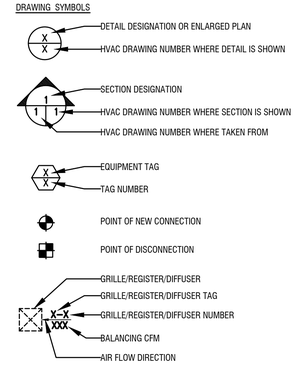
GENERAL PIPING NOTES:

1. INSTALL PIPING TO ALLOW ACCESS VALVES, AIR VENTS, EQUIPMENT REQUIRING ACCESS, AND TO PROVIDE MAXIMUM HEADROOM.
2. PROVIDE OFFSETS TO MAINTAIN CEILING HEIGHT AND TO COORDINATE WITH OTHER TRADES.
3. INSTALL VALVES IN HORIZONTAL PIPING WITH VALVE STEMS AT OR ABOVE THE PIPE CENTERLINE.
4. ARRANGE PIPING FOR VENTING OF AIR AND DRAINAGE OF ENTIRE SYSTEM.
5. INSTALL CONDENSATE DRAIN PIPING PITCHED AT 1/8" PER FOOT IN DIRECTION OF FLOW.

GENERAL DUCTWORK NOTES:

1. CHANGES IN SHAPE OR DIMENSION SHALL BE MADE WITH MAXIMUM TRANSITION OF 1 TO 7.
2. SEPARATE GALVANIZED SHEET METAL FROM ALUMINUM OR COPPER WITH LEAD OR FELT GASKETS.
3. PROVIDE SUPPLEMENTAL STIFFENING AND SUPPORTS TO DUCTS AND APPARATUS CASINGS TO PREVENT DRUMMING, SAGGING AND TO PROVIDE A STRUCTURALLY SOUND ASSEMBLY.
4. INSTALL DUCT FROM SHOWER EXHAUST GRILLES GRADING DOWN TO EXHAUST GRILLE, WITHOUT DIPS OR TRAPS.
5. PROVIDE OFFSETS AND TRANSITIONS TO COORDINATE WITH OTHER WORK.
6. PROVIDE DUCTWORK AND TRANSITIONS TO CONNECT DUCTWORK TO EQUIPMENT AND COILS.
7. INSTALL FLEXIBLE DUCTWORK IN A FULLY EXTENDED CONDITION WITHOUT SAGS AND KINKS.
8. INSTALL DUCT MOUNTED SMOKE DETECTORS IN ACCESSIBLE LOCATIONS.
9. UNLESS NOTED OTHERWISE, PROVIDE 1" THICK DUCT LINING FOR A MINIMUM OF 10 FEET OF DUCTWORK FROM THE SUPPLY AIR DISCHARGE AND RETURN AIR INLET OF AIR HANDLING UNITS, ENERGY RECOVERY UNITS, AND BLOWER COILS. FOR ALL LINED DUCTWORK, DIMENSIONS INDICATED ON DRAWINGS SHALL BE INSIDE CLEAR DIMENSIONS MEASURED FROM FACE-OF-LINER TO FACE-OF-LINER. LINING IS NOT REQUIRED FOR TOILET EXHAUST FANS. ROOF MOUNTED DUCTS ARE TO BE LINED AS DESCRIBED ABOVE AND ARE TO BE INSULATED WITH 2" THICK RIGID INSULATION AND WRAPPED WITH EPDM MATERIAL, SAME COLOR AS ROOF.
10. INSTALL DUCTS CONVEYING GREASE LADEN AIR AT A PITCH OF 1/4" PER FOOT OPPOSITE THE DIRECTION OF FLOW

DRAWING SYMBOLS



GENERAL AUTOMATIC TEMPERATURE CONTROLS NOTES:

1. TRANSFORMERS OR FILTERS FOR OPERATION OF AUTOMATIC TEMPERATURE CONTROLS FROM BUILDING POWER CIRCUITS SHALL BE PROVIDED UNDER DIVISION 23.
2. WIRING LOWER THAN 110 VOLTS FOR INTERLOCKED DEVICES, DDC CONTROLLERS, TERMINAL CONTROL UNITS, FLOW MEASURING DEVICES, AND OTHER POWER CONSUMING CONTROL DEVICES SHALL BE FURNISHED AND INSTALLED UNDER DIVISION 23. WIRING 110 VOLTS AND HIGHER FOR INTERLOCKED DEVICES, DDC CONTROLLERS, TERMINAL CONTROL UNITS, FLOW MEASURING DEVICES, AND OTHER POWER CONSUMING CONTROL DEVICES SHALL BE FURNISHED AND INSTALLED UNDER DIVISION 26.
3. PROVIDE SUPPLEMENTAL STIFFENING AND SUPPORTS TO DUCTS AND APPARATUS CASINGS TO PREVENT DRUMMING, SAGGING AND TO PROVIDE A STRUCTURALLY SOUND ASSEMBLY.
4. BRANCH CIRCUIT WIRING AND CONDUIT FURNISHED FOR CONTROL EQUIPMENT POWER SHALL BE SEPARATE FROM OTHER POWER WIRING. EACH CIRCUIT SHALL EXTEND TO A 120V BRANCH CIRCUIT PANEL, AND IDENTIFIED 120V, 20 AMPERE, SINGLE POLE BRANCH CIRCUIT BREAKER FURNISHED IN THE PANEL TO SERVE THE CIRCUIT. NO MORE THAN 2 DDC CONTROLLER INSTALLATIONS SHALL OPERATE FROM A SINGLE 120V BRANCH CIRCUIT, UNLESS INDICATED OTHERWISE.
5. WHERE SYSTEMS ARE SERVED BY EMERGENCY POWER, CONTROLS FOR OPERATION OF THOSE SYSTEMS SHALL ALSO BE SERVED BY EMERGENCY POWER.
6. WHERE DAMPERS PREVENT AIRFLOW THROUGH AN AIR HANDLING UNIT OR FAN, THOSE DAMPERS SHALL BE PROVEN OPEN PRIOR TO STARTING THE UNIT OR FAN. PROOF SHALL BE BY MECHANICAL SAFETY LIMIT SWITCH ACTIVATED BY THE DAMPER BLADE. FOR SERVICE WITH VARIABLE FREQUENCY DRIVES THE SWITCH SHALL BE WIRED IN THE AUTOMATIC AND HAND/TEST POSITIONS AND IN THE BYPASS POSITION FOR VARIABLE FREQUENCY DRIVES WITH BYPASS.
7. ALL LOW VOLTAGE WIRING AND AIR PIPING OR TUBING SHALL BE PLENUM RATED. MECHANICAL CONTRACTOR SHALL FURNISH ALL LOW VOLTAGE WIRING, AIR PIPING, AND TUBING REQUIRED FOR AUTOMATIC TEMPERATURE CONTROLS SYSTEMS. LOW VOLTAGE WIRING IS ALL WIRE OPERATING AT A VOLTAGE LOWER THAN 110 VOLTS.
8. ALL TEMPERATURE CONTROL SHALL HAVE A 5 DEGREE DEAD-BAND WITH OVERLAP RESTRICTIONS. EQUIPMENT SHALL BE PROVIDED WITH AT LEASE ONE MEANS OF EMERGENCY SHUT DOWN. SET BACK CONTROL SHALL ALLOW FOR AUTOMATIC RESTART AS WELL AS TEMPORARY OPERATION AS REQUIRED BY MAINTENANCE.

GENERAL MECHANICAL NOTES:

1. CODES AND STANDARDS LISTED IN SPECIFICATIONS AND DRAWINGS ARE MINIMUM STANDARDS. WHERE REQUIREMENTS ON THE DRAWINGS OR SPECIFICATIONS EXCEED THE MINIMUM CODE REQUIREMENTS, THE DRAWINGS OR SPECIFICATIONS SHALL GOVERN.
2. THE POWER RATING OF MOTORS AND OTHER MECHANICAL EQUIPMENT AND THE ELECTRICAL CHARACTERISTICS OF ELECTRICAL SYSTEMS SERVING THEM HAVE BEEN ESTABLISHED AS MINIMUMS WHICH ALLOW THAT EQUIPMENT TO FUNCTION PROPERLY TO PRODUCE THE REQUIRED CAPACITIES. POWER RATINGS INCLUDE REASONABLE SAFETY FACTORS TO ACCOMMODATE COMMON DIFFERENCES BETWEEN DESIGN PARAMETERS AND FIELD CONSTRUCTION PRACTICES. EQUIPMENT WITH POWER RATINGS LESS THAN THOSE INDICATED ON THE DRAWINGS SHALL NOT BE PERMITTED.
3. REASONABLE EFFORTS HAVE BEEN MADE TO COORDINATE ELECTRICAL REQUIREMENTS OF MECHANICAL EQUIPMENT WITH THE ELECTRICAL SYSTEMS SERVING THAT EQUIPMENT. DIFFERENCES AMONG MANUFACTURERS OF MECHANICAL EQUIPMENT MAKE IT IMPOSSIBLE TO PRODUCE A SINGLE ELECTRICAL DESIGN WHICH WILL SATISFY THE VARYING ELECTRICAL REQUIREMENTS OF THE THOSE MANUFACTURERS. CONSEQUENTLY, THE CONTRACTOR SHALL COORDINATE THE ELECTRICAL REQUIREMENTS OF THE MECHANICAL EQUIPMENT ACTUALLY FURNISHED ON THIS PROJECT WITH THE EQUIPMENT ACTUALLY FURNISHED ON THIS PROJECT AND PROVIDE ELECTRICAL SYSTEMS REQUIRED BY THAT EQUIPMENT. THIS COORDINATION EFFORT SHALL BE COMPLETED PRIOR TO THE INSTALLATION OF EITHER THE MECHANICAL EQUIPMENT OR THE ELECTRICAL SYSTEMS SERVING THAT EQUIPMENT. ELECTRICAL SYSTEM REVISIONS REQUIRED TO COORDINATE WITH THE MECHANICAL EQUIPMENT ACTUALLY FURNISHED SHALL BE PROVIDED AT NO ADDITIONAL COST TO THE OWNER.
4. DRAWINGS INDICATE GENERAL LOCATIONS OF FIXTURES, APPARATUS, EQUIPMENT, PIPING, AND DUCTWORK. CHANGES ON LOCATION SHALL BE MADE TO ACCOMMODATE EXISTING OR NEW BUILDING CONDITIONS AND COORDINATION WITH OTHER TRADES, INCLUDING HVAC, PLUMBING, ELECTRICAL, FIRE PROTECTION, STRUCTURAL, AND ARCHITECTURAL, SHALL BE MADE WITHOUT ADDITIONAL COST TO THE OWNER.
5. THOROUGHLY CLEAN/FLUSH EXISTING AND NEW HYDRONIC PIPING SYSTEMS WITH CLEAN WATER. AFTERWARDS, REMOVE AND CLEAN OR REPLACE STRAINER SCREENS.
6. ALL HVAC SYSTEMS SHALL BE TESTED AND BALANCED ACCORDING TO NEBB AND SMACNA STANDARDS. PROVIDE REPORT UPON COMPLETION.
7. PROVIDE ACCESS TO EQUIPMENT AND PORTIONS OF BUILDING SYSTEMS REQUIRING SERVICE.
8. DO NOT INSTALL DUCTWORK, PIPING, OR EQUIPMENT IN ELECTRICAL ROOMS, ELEVATOR ROOMS, OR ELEVATOR SHAFTS, UNLESS EXPLICITLY INDICATED ON THE DRAWINGS. PIPING, DUCTWORK, AND EQUIPMENT (SWITCHGEAR, SWITCHBOARDS, PANELS, MOTOR CONTROL CENTERS, VARIABLE FREQUENCY DRIVES, TRANSFORMERS, OR STARTERS) SHALL NOT BE INSTALLED DIRECTLY ABOVE OR 42" IN FRONT OF ELECTRICAL EQUIPMENT FROM THE FLOOR TO THE STRUCTURE ABOVE.
9. PROVIDE START UP AND COMMISSIONING OF ALL EQUIPMENT PROVIDED IN COMPLIANCE WITH THE MANUFACTURERS WRITTEN INSTRUCTIONS. PROVIDE START UP AND WARRANTEE PAPERWORK AT THE COMPLETION OF WORK. WORK SHALL BE COMPLETED BY THE MANUFACTURER OR A MANUFACTURERS' CERTIFIED FIRM OR TECHNICIAN. CONFIRM CALIBRATION OF ALL SENSORS AND ADJUST AS REQUIRED.
10. UNLESS INDICATED OTHERWISE, EQUIPMENT AND MATERIALS SHALL BE NEW AND OF THE CUSTOMARY STANDARD AND QUALITY FURNISHED BY THE DESIGNATED MANUFACTURER FOR THAT CATALOG NUMBER.
11. AIR SYSTEMS SHALL OPERATE WITHOUT AERODYNAMIC NOISE GENERATED FROM FAULTY INSTALLATION OF DUCTWORK, DIFFUSERS, OR ANY PORTION OF THE AIR DISTRIBUTION SYSTEM.
12. SUPPORT PIPING INDEPENDENTLY OF EQUIPMENT. HANGER RODS SHALL BE SUSPENDED FROM THE STRUCTURE. DO NOT SUSPEND FROM OTHER PIPING, CONDUIT, EQUIPMENT, OR DUCTWORK.
13. ALL WORK REFERENCED UNDER DIVISION 23 SHALL BE DONE BY THE MECHANICAL CONTRACTOR.
14. DRAWINGS INDICATE DESIGN INTENT. CONTRACTOR IS RESPONSIBLE TO FIELD VERIFY ALL INSTALLATIONS. CONTRACTOR IS RESPONSIBLE FOR COORDINATION BETWEEN OTHER TRADES TO ASSURE THE PROPER INSTALLATION OF ALL EQUIPMENT.
15. ALL PIPING, DUCTWORK, INSULATION, CONDUITS, SUPPORTS AND HVAC EQUIPMENT EXPOSED TO VEIW SHALL BE PAINTED. COLOR SHALL BE SELECED BY ARCHITECT.
16. WHERE DUCTWORK IS EXPOSED DUCT SEAMS SHALL BE MINIMIZED AND SHALL BE OF HIGH QUALITY WORKMANSHIP. ALL DUCTWORK SHALL BE CONSTRUCTED AND SEALED IN ACCORDANCE WITH SMACNA STANDARDS.
17. ALL MATERIALS EXPOSED WITHIN THE PLENUM SHALL BE NON COMBUSTIBLE OR SHALL HAVE A FLAME SPREAD INDEX OF NOT MORE THAN 25 AND A SMOKE-DEVELOPED INDEX OF NOT MORE THAN 50 WHEN TESTED IN ACCORDANCE WITH ASTM E-84.

ADDITIONAL MECHANICAL REQUIREMENTS:

1. DRAWINGS ARE SCHEMATIC IN NATURE INTENDED TO EXEMPLIFY CODE COMPLIANCE FOR THE PURPOSE OF OBTAINING A CONSTRUCTION PERMIT. THE CONTRACTOR SHALL ASSURE THE PROPER INSTALLATION AND OPERATION OF ALL ASSOCIATED SYSTEMS.
2. THE INSTALLATION AND MATERIALS SHALL BE IN ACCORDANCE WITH ALL LOCAL, STATE AND FEDERAL CODES.
3. PROVIDE R-5 (INSTALLED VALUE) DUCTWORK INSULATION WITH VAPOR BARRIER IN INTERIOR SPACES. INSTALL PER THE MANUFACTURER'S WRITTEN INSTRUCTIONS. INSULATION SHALL BE PROVIDED ON RETURN AIR SYSTEMS WHERE THE DUCTWORK IS NOT LOCATED WITHIN CONDITIONED SPACES. WHERE DUCTWORK IS INSTALLED OUTSIDE, PROVIDE R-8 BOARD WITH WEATHER PROOF JACKET. MATERIALS SHALL BE COMPLIANCE WITH ALL APPLICABLE ASTM TESTS AS WELL AS NFPA 90A AND 90B.
4. DUCTWORK SHALL BE GALVANIZED SHEET STEEL IN THE GAUGE AS REQUIRED PER THE LATEST VERSION OF SMACNA GUIDELINES.
5. PROVIDE SUPPLEMENTAL STIFFENING AND SUPPORTS TO DUCTS AND APPARATUS CASINGS TO PREVENT DRUMMING, SAGGING AND TO PROVIDE A STRUCTURALLY SOUND ASSEMBLY.
6. PROVIDE ALL DUCTWORK FITTINGS INCLUDING BUT NOT LIMITED TO TEES, TAPS, ELBOWS, VOLUME DAMPERS ETC IN ACCORDANCE WITH THE LATEST VERSION OF SMACNA GUIDELINES.
7. COORDINATE ELECTRICAL POWER REQUIREMENTS WITH THE ELECTRICAL CONTRACTOR. PROVIDE MEANS OF DISCONNECT FOR EQIPMENT AS REQUIRED.
8. THE CONTRACTOR SHALL ADJUST DUCTWORK AND EQUIPMENT LAYOUT IN FIELD AS REQUIRED TO FACILITATE A NEAT AND HIGH QUALITY INSTALLATION.
9. PROVIDE CONTROL WIRING AND DEVICES IN COMPLIANCE WITH THE CURRENTLY ADOPTED VERSION OF THE NATIONAL ELECTRIC CODE.
10. DO NOT INSTALL SERVICEABLE EQUIPMENT WITHIN 10' OF ROOF EDGES
11. DO NOT INSTALL AIR INTAKES WITHIN 10' OF EXHAUST TERMINALS OR PLUMBING VENTS
12. FURNISH IOM MANUALS AND AS-BUILT DRAWINGS WITH 90 DAYS OF COMPLETION OF WORK
13. ALL EXPOSED DUCTWORK SHALL BE DOUBLE WALLED OR INTERNALLY LINED TO PREVENT THE FORMATION OF CONDENSATION.
14. ALL REFRIGERATION PIPING SHALL BE ACR TYPE COPPER TUBE WITH BRAZED FITTINGS. SIZED IN ACCORDANCE WITH ASSOCIATED EQUIPMENT MANUFACTURER'S WRITTEN INSTRUCTIONS. INSULATE ALL REFRIGERANT PIPING WITH 1" FLEXIBLE ELASTOMERIC LINER. PROVIDE INSULATION MANUFACTURER'S WEATHER-PROOF MASTIC FOR ALL OUTDOOR INSTALLATIONS.

PROJECT NAME / LOCATION:

PROPOSED "INTERIOR AND EXTERIOR ALTERATIONS"

India Pentecostal Assembly

310A SOUTH OYSTER BAY ROAD
SYOSSET, NY 11791

TOWN OF OYSTER BAY

ARCHITECT:

the KENT ARCHITECTURAL COLLABORATIVE, PLLC
132 CLYDE STREET, SUITE 1B
WEST SAYVILLE • NEW YORK • 11796
631.793.9684
rkent.tkac@gmail.com

CONSTRUCTION MANAGER / GENERAL CONTRACTOR:

CCC CORE CONCEPTS CONTRACTING
687 OLD WILLETS PATH, HAUPPAUGE, NY 11788
www.coreconceptscontracting.com 631.435.8985

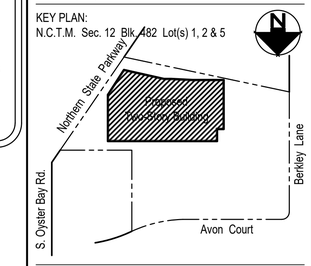
STRUCTURAL ENGINEER:

reuther&bowen Engineering, Design, Construction Services
326 WARD ST. SCRANTON, PA 18512-2424
PHONE (570) 496-7020 FAX (570) 496-7021

MECHANICAL / ELECTRICAL / PLUMBING ENGINEERS:

DMG designmanagementgroup building systems, smarter
2007 HIGHWAY 315, SUITE 202 PITTSTON, PA 18640
PHONE (570) 299-7520

KEY PLAN:
N.C.T.M. Sec. 12 Blk. 482 Lot(s) 1, 2 & 5



REVISION:	DATE:
ISSUED FOR CONSTRUCTION CORE & SHELL	2025.03.05
REV PER TOOB COMMENTS	2024.11.22
ISSUED FOR PERMIT	2024.07.31

SEAL:

DRAWING TITLE:
MECHANICAL COVER SHEET

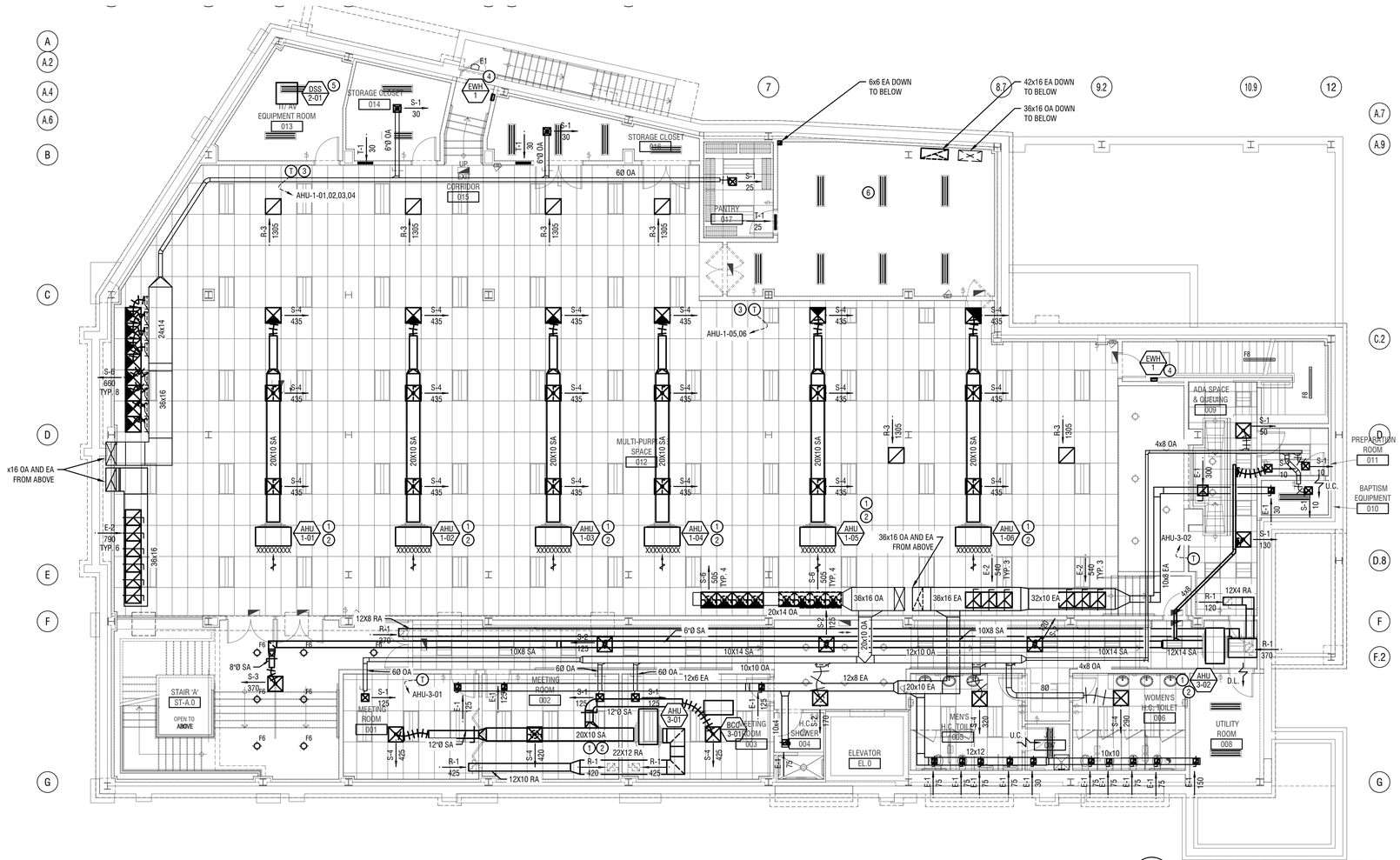
DATE: 2024.01.29

DRAWN BY: RK/BMF

PROJECT NO.: 24.03

DRAWING NO.: M0.1

TOB APPLICATION No. 22030355



1 M2.1 MECHANICAL PLAN - BASEMENT

SCALE: 1/8" = 1'-0"

KEYED CONSTRUCTION NOTES

1. SPLIT SYSTEM VRF AIR HANDLER MOUNTED ABOVE CEILING. HANG FROM STRUCTURE WITH VIBRATION ISOLATION. INSTALL PER MANUFACTURER'S WRITTEN INSTRUCTIONS. COORDINATE FINAL LOCATION WITH ARCHITECT AND BUILDING OWNER PRIOR TO CONSTRUCTION. SEE PIPING PLAN, SCHEDULE AND DETAILS FOR MORE INFORMATION.
2. DISCHARGE CONDENSATE AT NEAREST APPROVED LOCATION. PROVIDE CONDENSATE PUMP AS REQUIRED.
3. WALL MOUNTED CONTROLLER FOR VRF AIR HANDLING UNIT. COORDINATE FINAL LOCATION WITH OWNER PRIOR TO CONSTRUCTION. WIRE BACK TO UNIT AS SHOWN.
4. ELECTRIC WALL HEATER WITH INTEGRAL TAMPERPROOF THERMOSTAT. PROVIDE SURFACE MOUNTING FRAME. INSTALL PER MANUFACTURER'S WRITTEN INSTRUCTIONS. SEE SCHEDULE FOR MORE INFORMATION. COORDINATE INSTALLATION WITH OWNER, ARCHITECT, AND ALL OTHER TRADES.
5. CEILING MOUNTED DUCTLESS SPLIT SYSTEM. PROVIDE DISCONNECT SWITCH AND ALL NECESSARY MOUNTING HARDWARE. PROVIDE CONDENSATE PUMP AND DISCHARGE TO NEAREST APPROVED LOCATION. INSTALL PER MANUFACTURER'S WRITTEN INSTRUCTIONS. SEE EQUIPMENT SCHEDULE FOR MORE INFORMATION. WIRE TO THERMOSTAT SHOWN. COORDINATE FINAL LOCATION OF UNIT AND THERMOSTAT WITH ARCHITECT AND BUILDING OWNER.
6. FINAL KITCHEN DESIGN FOR COOKING HOOD, DISHWASHER HOOD, MAKEUP AIR UNIT, AND EXHAUST FANS SHALL BE BY QUALIFIED KITCHEN AIR CONSULTANT.

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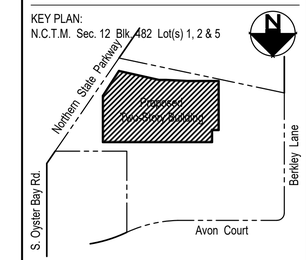

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REVISION:	DATE:

SEAL:

STATE OF NEW YORK · RICHARD DENNIS REEDY · 097739 · LICENSED PROFESSIONAL ENGINEER

DRAWING TITLE:
MECHANICAL DUCTWORK PLAN - BASEMENT

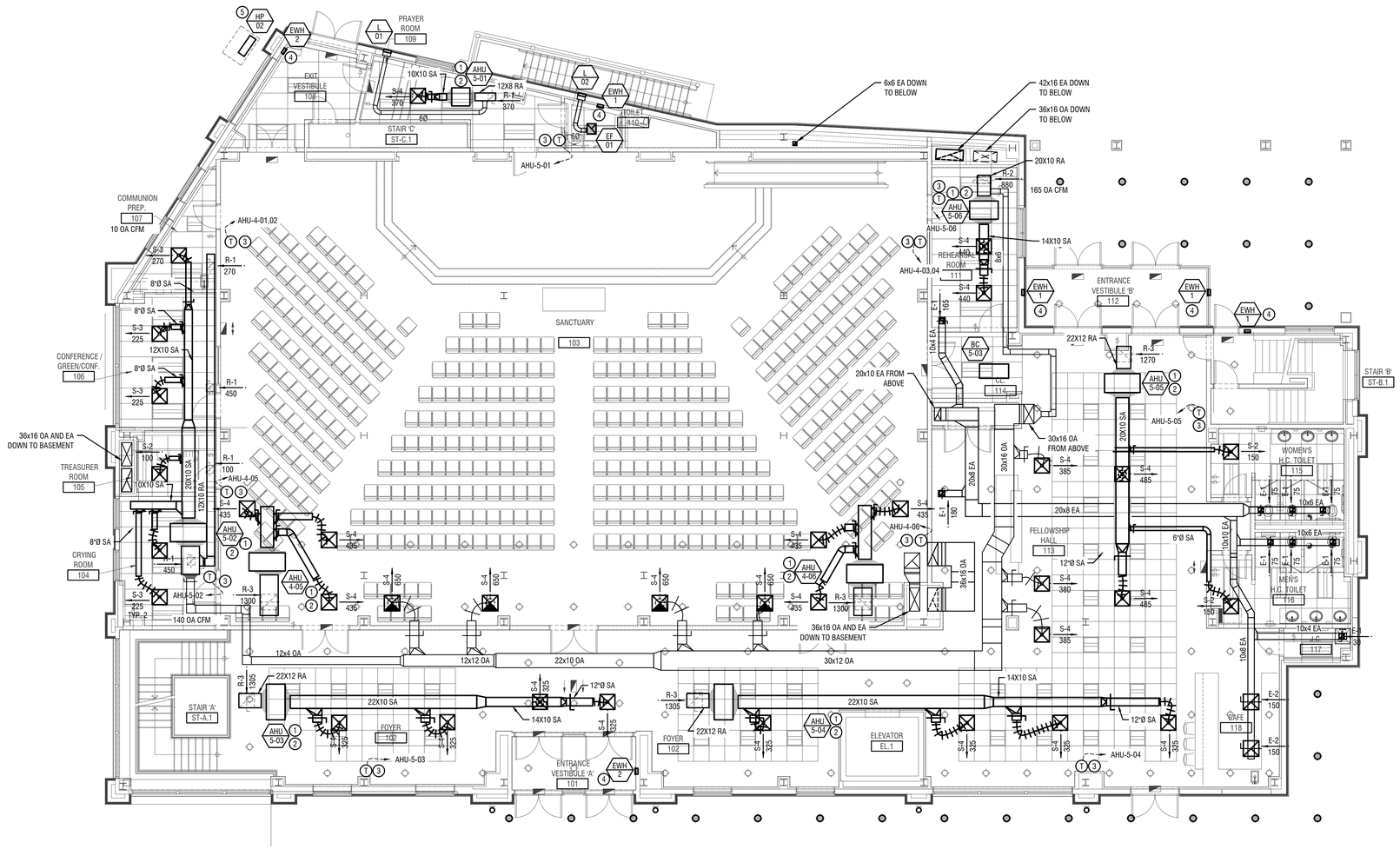
DATE: 2024.01.29

DRAWN BY: RK/BMF

PROJECT NO.: 24.03

DRAWING NO.: M2.1

TOB APPLICATION No. 22030355



1 M2.2 MECHANICAL PLAN - FIRST FLOOR

SCALE: 1/8" = 1'-0"

KEYED CONSTRUCTION NOTES

1. SPLIT SYSTEM VRF AIR HANDLER MOUNTED ABOVE CEILING. HANG FROM STRUCTURE WITH VIBRATION ISOLATION. INSTALL PER MANUFACTURER'S WRITTEN INSTRUCTIONS. COORDINATE FINAL LOCATION WITH ARCHITECT AND BUILDING OWNER PRIOR TO CONSTRUCTION. SEE PIPING PLAN, SCHEDULE AND DETAILS FOR MORE INFORMATION.
2. DISCHARGE CONDENSATE AT NEAREST APPROVED LOCATION. PROVIDE CONDENSATE PUMP AS REQUIRED.
3. WALL MOUNTED CONTROLLER FOR VRF AIR HANDLING UNIT. COORDINATE FINAL LOCATION WITH OWNER PRIOR TO CONSTRUCTION. WIRE BACK TO UNIT AS SHOWN.
4. ELECTRIC WALL HEATER WITH INTEGRAL TAMPERPROOF THERMOSTAT. PROVIDE SURFACE MOUNTING FRAME. INSTALL PER MANUFACTURER'S WRITTEN INSTRUCTIONS. SEE SCHEDULE FOR MORE INFORMATION. COORDINATE INSTALLATION WITH OWNER, ARCHITECT, AND ALL OTHER TRADES.
5. MOUNT OUTDOOR HEAT PUMP ON MINI SPLIT STANDS.FOLLOWING ALL MANUFACTURER'S WRITTEN INSTRUCTIONS. ROUTE REFRIGERANT PIPING THRU WALL TO INDOOR UNIT. SEAL ALL EXTERIOR PENETRATIONS WEATHER-TIGHT. ROUTE AND INSULATE PIPING FOLLOWING MANUFACTURER'S RECOMMENDATIONS. SEE EQUIPMENT SCHEDULE FOR MORE INFORMATION.

PROJECT NAME / LOCATION:

PROPOSED "INTERIOR AND EXTERIOR ALTERATIONS"

India Pentecostal Assembly

310A SOUTH OYSTER BAY ROAD
SYOSSET, NY 11791

TOWN OF OYSTER BAY

ARCHITECT:

the KENT ARCHITECTURAL COLLABORATIVE, PLLC
132 CLYDE STREET, SUITE 1B
WEST SAYVILLE • NEW YORK • 11796
631.793.9684
rkent.tkac@gmail.com

CONSTRUCTION MANAGER / GENERAL CONTRACTOR:

CCC CORE CONCEPTS CONTRACTING
687 OLD WILLETS PATH, HAUPPAUGE, NY 11788
www.coreconceptscontracting.com 631.435.8985

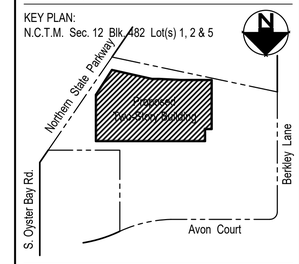
STRUCTURAL ENGINEER:

reuther&bowen Engineering, Design, Construction Services
326 WARD ST. SCRANTON, PA 18512-2424
PHONE (570) 496-7020 FAX (570) 496-7021

MECHANICAL / ELECTRICAL / PLUMBING ENGINEERS:

DMG designmanagementgroup building systems, smarter
2007 HIGHWAY 315, SUITE 202 PITTSTON, PA 18640
PHONE (570) 299-7520

KEY PLAN:
N.C.T.M. Sec. 12 Blk. 482 Lot(s) 1, 2 & 5



REVISION:	DATE:
ISSUED FOR CONSTRUCTION CORE & SHELL	2025.03.05
REV PER TOOB COMMENTS	2024.11.22
ISSUED FOR PERMIT	2024.07.31

SEAL:

STATE OF NEW YORK · RICHARD DENNIS BERNOTT · 097739 · LICENSED PROFESSIONAL ENGINEER

DRAWING TITLE:
MECHANICAL DUCTWORK PLAN - FIRST FLOOR

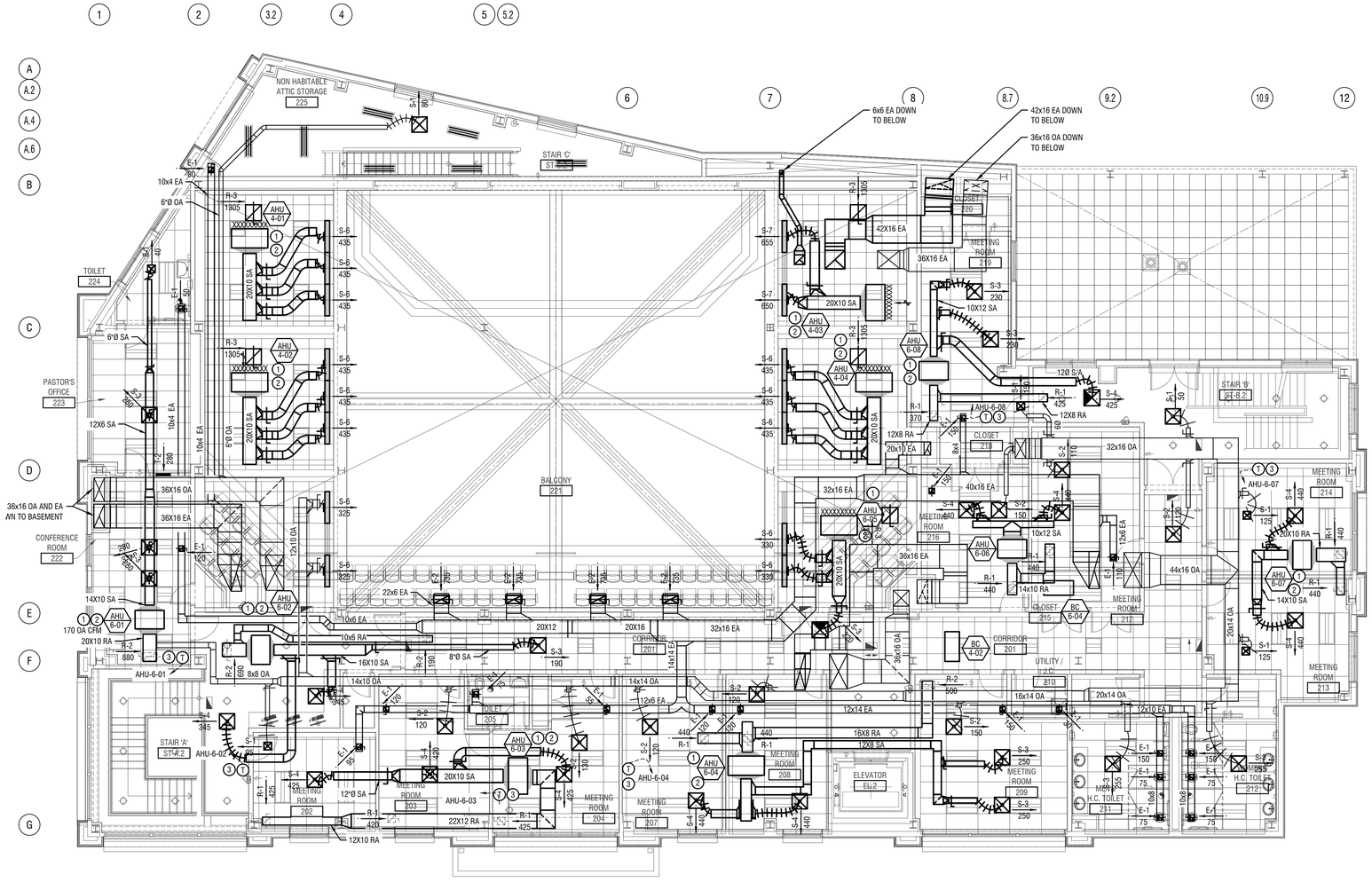
DATE: 2024.01.29

DRAWN BY: RK/BMF

PROJECT NO.: 24.03

DRAWING NO.: M2.2

TOB APPLICATION No. 22030355



1 M2.3 MECHANICAL PLAN - SECOND FLOOR

SCALE: 1/8" = 1'-0"

KEYED CONSTRUCTION NOTES

1. SPLIT SYSTEM VRF AIR HANDLER MOUNTED ABOVE CEILING. HANG FROM STRUCTURE WITH VIBRATION ISOLATION. INSTALL PER MANUFACTURER'S WRITTEN INSTRUCTIONS. COORDINATE FINAL LOCATION WITH ARCHITECT AND BUILDING OWNER PRIOR TO CONSTRUCTION. SEE PIPING PLAN, SCHEDULE AND DETAILS FOR MORE INFORMATION.
2. DISCHARGE CONDENSATE AT NEAREST APPROVED LOCATION. PROVIDE CONDENSATE PUMP AS REQUIRED.
3. WALL MOUNTED CONTROLLER FOR VRF AIR HANDLING UNIT. COORDINATE FINAL LOCATION WITH OWNER PRIOR TO CONSTRUCTION. WIRE BACK TO UNIT AS SHOWN.
4.
5.
6.

PROJECT NAME / LOCATION:

PROPOSED "INTERIOR AND EXTERIOR ALTERATIONS"

India Pentecostal Assembly

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WEST SAYVILLE • NEW YORK • 11796
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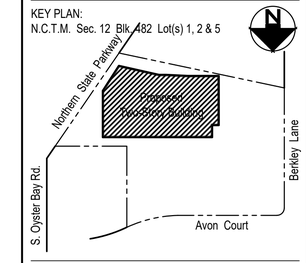
KEY PLAN:
N.C.T.M. Sec. 12 Blk. 482 Lot(s) 1, 2 & 5



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REV PER TOOB COMMENTS	2024.11.22
ISSUED FOR PERMIT	2024.07.31

SEAL:

STATE OF NEW YORK · RICHARD DENNIS REBER · 097739 · LICENSED PROFESSIONAL ENGINEER

DRAWING TITLE:
MECHANICAL DUCTWORK PLAN - SECOND FLOOR

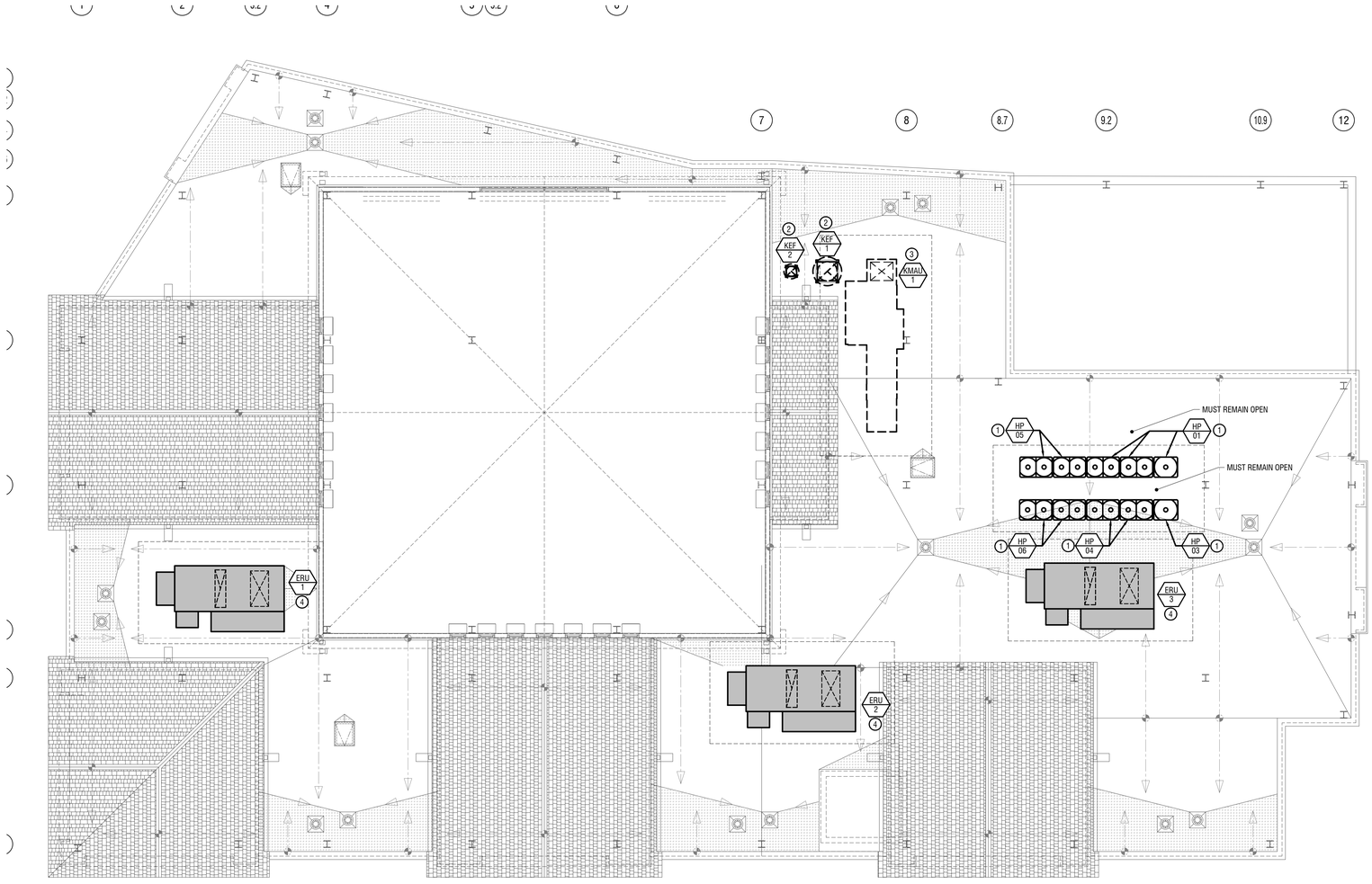
DATE: 2024.01.29

DRAWN BY: RK/BMF

PROJECT NO.: 24.03

DRAWING NO.: M2.3

TOB APPLICATION No. 22030355



1 / M2.4 MECHANICAL PLAN - ROOF PLAN

SCALE: 1/8" = 1'-0"

KEYED CONSTRUCTION NOTES

1. MOUNT OUTDOOR HEAT PUMP ON ROOF ON MINI SPLIT STANDS FOLLOWING ALL MANUFACTURER'S WRITTEN INSTRUCTIONS. ROUTE REFRIGERANT PIPING THRU ROOF TO INDOOR UNIT PROVIDING PIPE PORTAL AT ROOF PENETRATION. SEAL ALL EXTERIOR PENETRATIONS WEATHER-TIGHT. ROUTE AND INSULATE PIPING FOLLOWING MANUFACTURER'S RECOMMENDATIONS. SEE EQUIPMENT SCHEDULE FOR MORE INFORMATION. MAINTAIN PROPER SERVICE CLEARANCES AS PER MANUFACTURER'S WRITTEN INSTRUCTIONS.
2. PROVIDE NEW KITCHEN EXHAUST FAN LOCATED ON ROOF. FAN TO BE INTERLOCKED WITH ASSOCIATED KITCHEN HOOD. MOUNT FAN ON ROOF CURB AS SCHEDULED. WIRE TO HOOD CONTROL PANEL AS INDICATED ON PLAN.
3. PROVIDE NEW MAKE-UP AIR UNIT. UNIT TO BE INTERLOCKED WITH TEMPERATURE AND STATIC PRESSURE SENSORS AS SHOWN ON PLANS.
4. NEW ENERGY RECOVERY UNIT AND CURB. INSTALL PER MANUFACTURER'S WRITTEN INSTRUCTIONS.

PROJECT NAME / LOCATION:

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India Pentecostal Assembly

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the KENT ARCHITECTURAL COLLABORATIVE, PLLC
132 CLYDE STREET, SUITE 1B
WEST SAYVILLE • NEW YORK • 11796
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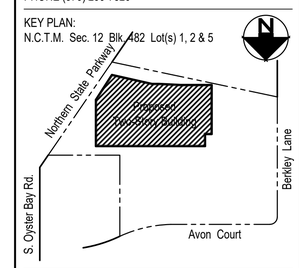

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REV PER TOOB COMMENTS	2024.11.22
ISSUED FOR PERMIT	2024.07.31

SEAL:

DRAWING TITLE:
MECHANICAL PLAN - ROOF PLAN

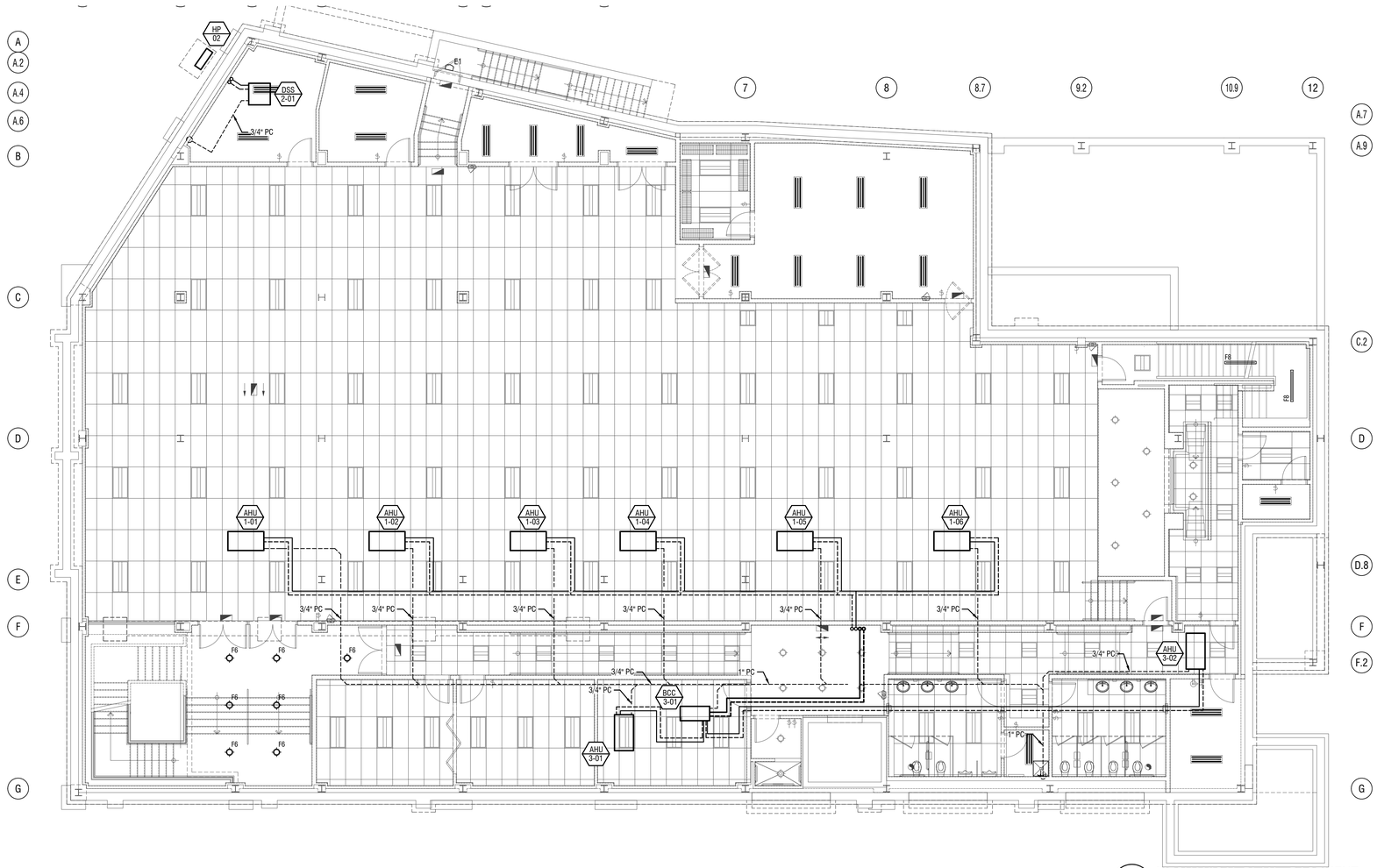
DATE: 2024.01.29

DRAWN BY: RK/BMF

PROJECT NO.: 24.03

DRAWING NO.: M2.4

TOB APPLICATION No. 22030355



1 M3.1 MECHANICAL PLAN - BASEMENT

SCALE: 1/8" = 1'-0"

PROJECT NAME / LOCATION:

PROPOSED "INTERIOR AND EXTERIOR ALTERATIONS"

India Pentecostal Assembly

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SYOSSET, NY 11791

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132 CLYDE STREET, SUITE 1B
WEST SAYVILLE • NEW YORK • 11796
631.793.9684
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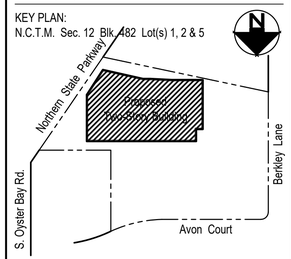

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REV PER TOOB COMMENTS	2024.11.22
ISSUED FOR PERMIT	2024.07.31

SEAL:

STATE OF NEW YORK · RICHARD DENNIS REEDER · 097739 · LICENSED PROFESSIONAL ENGINEER

DRAWING TITLE:
MECHANICAL PIPING PLAN - BASEMENT

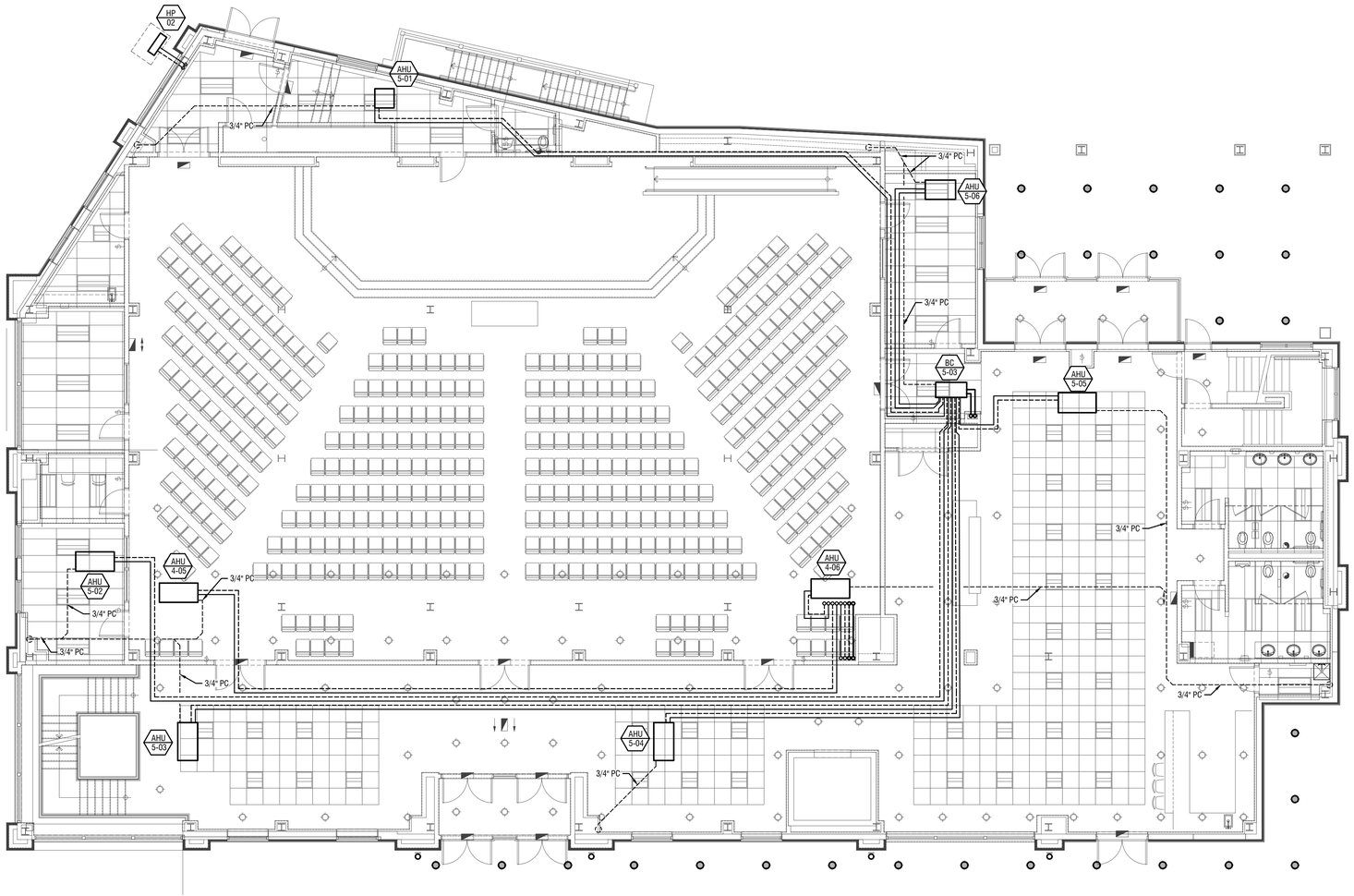
DATE: 2024.01.29

DRAWN BY: RK/BMF

PROJECT NO.: 24.03

DRAWING NO.: M3.1

TOB APPLICATION No. 22030355



1 / M3.2 MECHANICAL PLAN - FIRST FLOOR

SCALE: 1/8" = 1'-0"

PROJECT NAME / LOCATION:

PROPOSED "INTERIOR AND EXTERIOR ALTERATIONS"

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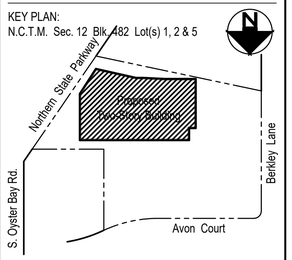

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ISSUED FOR PERMIT	2024.07.31

SEAL:

DRAWING TITLE:

MECHANICAL PIPING PLAN - FIRST FLOOR

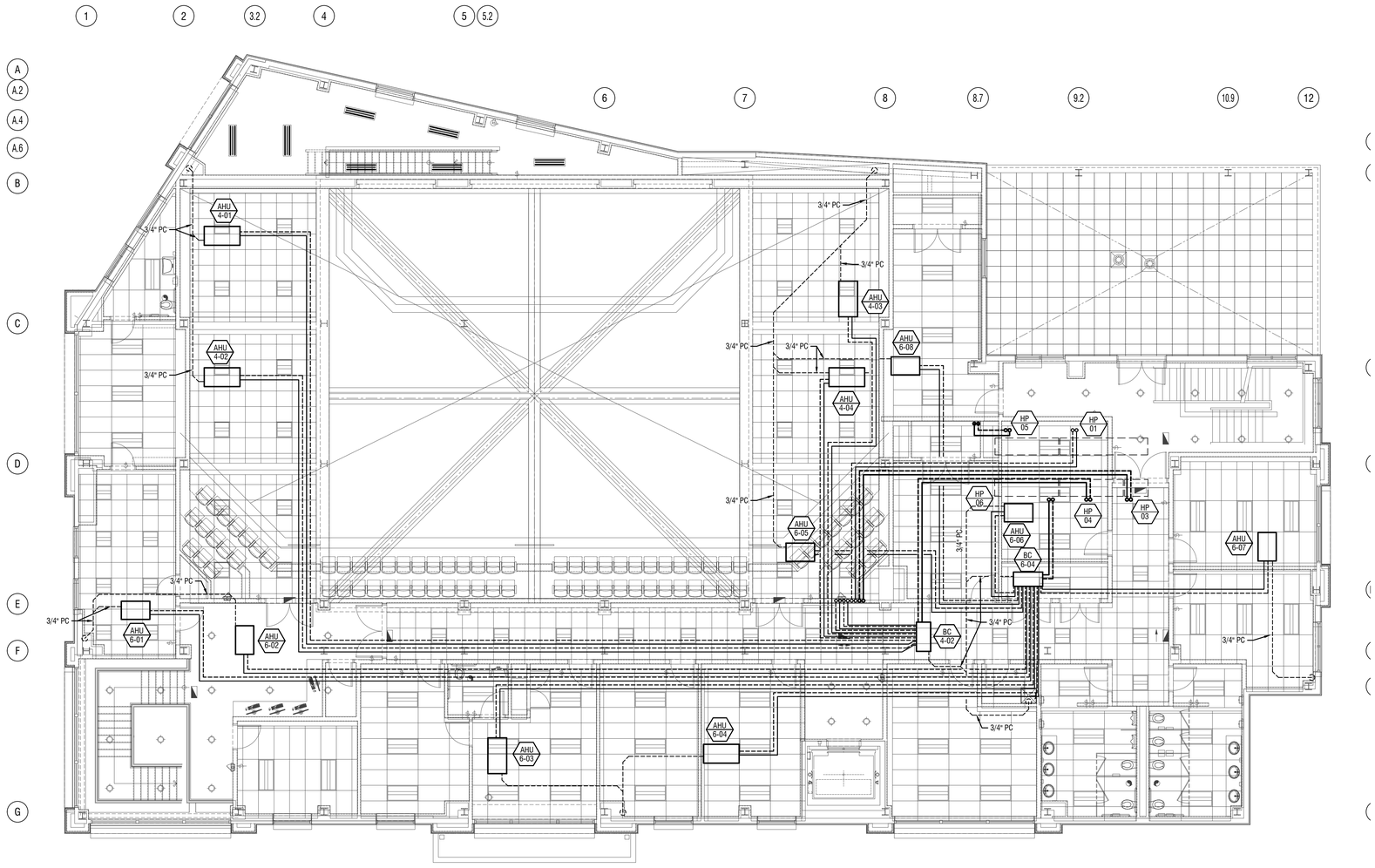
DATE: 2024.01.29

DRAWN BY: RK/BMF

PROJECT NO.: 24.03

DRAWING NO.: M3.2

TOB APPLICATION No. 22030355



1 M3.3 MECHANICAL PLAN - SECOND FLOOR

SCALE: 1/8" = 1'-0"

PROJECT NAME / LOCATION:

PROPOSED "INTERIOR AND EXTERIOR ALTERATIONS"

India Pentecostal Assembly

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KEY PLAN:
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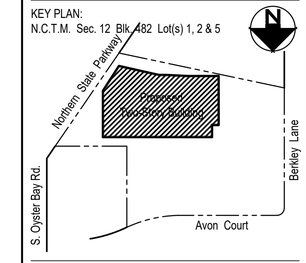

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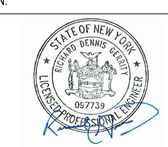


DRAWING TITLE:

MECHANICAL PIPING PLAN - SECOND FLOOR

DATE: 2024.01.29

DRAWN BY: RK/BMF

PROJECT NO.: 24.03

DRAWING NO.: M3.3

TOB APPLICATION No. 22030355

Quick Results

Connection Summary

Indoor Units:	6 / 2 to 50	
Capacity:	288 / 144 to 374 (100.0%)	
* Connectable capacity is not actual capacity.		
Total Pipe Length:	431.2 / 3280.0	feet
Furthest Actual:	70.0 / 541.0	feet
Furthest Equiv.:	106.8 / 623.0	feet
After 1st Branch Actual:	50.0 / 295.0	feet
After 1st Branch Equiv.:	59.2 / 295.0	feet

Refrigerant

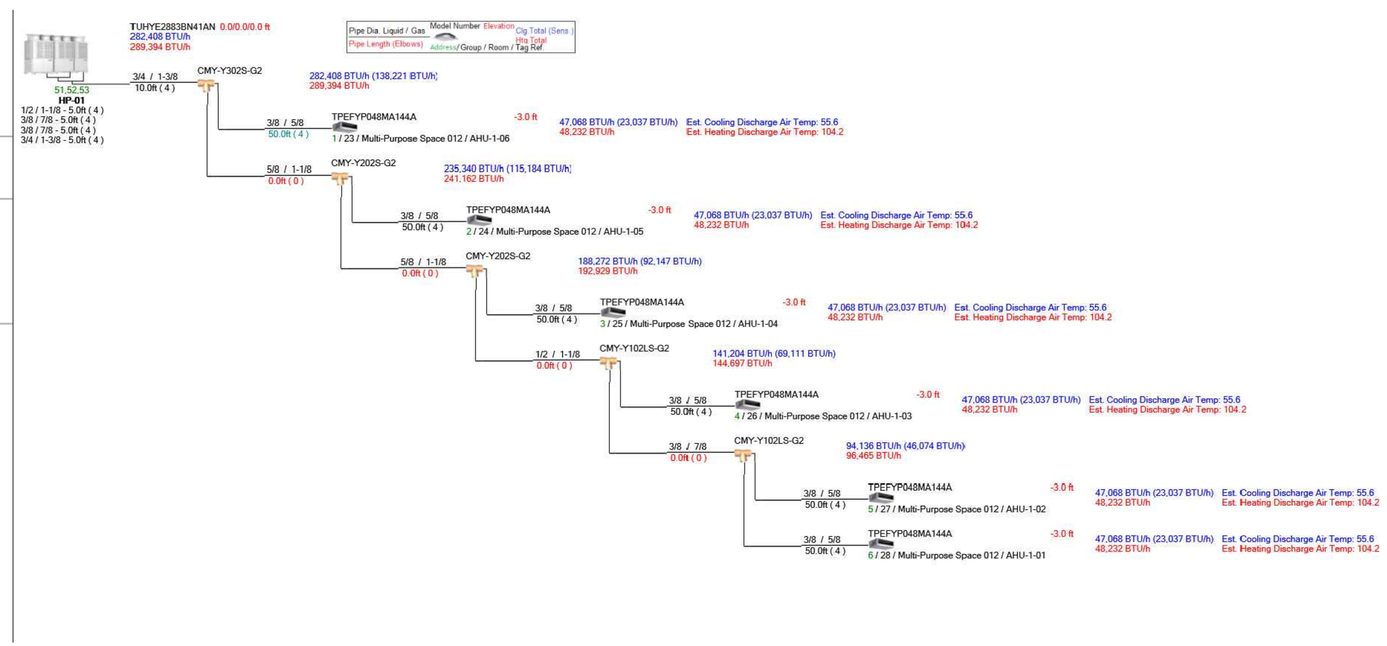
System Refrigerant:	**R410A**	
Additional Field-Charge:	**32.12**	**lb**
Total System Charge:	**89.66**	**lb**

Correction Factors

	Cooling	Heating
Outdoor Unit Capacity:	1.00	1.00
Outdoor Temperature:	1.03	0.96
Piping Length:	0.96	0.99
Defrosting:	-	0.95
User Derate:	1.00	1.00
Outdoor Altitude:	1.00	1.00
	0.98	**0.90**

Conditions

Cooling		Heating	
Indoor Dry Bulb	72.0 °F	Indoor Dry Bulb	70.0 °F
Indoor Wet Bulb	67.0 °F		
Indoor Humidity	77.8 %	Outdoor Dry Bulb	12.0 °F
		Outdoor Wet Bulb	11.3 °F
Outdoor Dry Bulb	90.0 °F	Outdoor Humidity	85.2 %



1 / M5.1 VRF PIPING DIAGRAM- HP-01

SCALE: NTS

PROJECT NAME / LOCATION:

PROPOSED "INTERIOR AND EXTERIOR ALTERATIONS"

India Pentecostal Assembly

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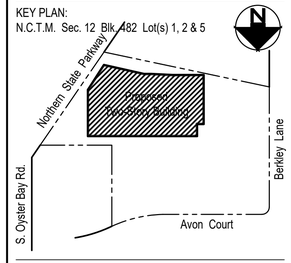


ISSUED FOR CONSTRUCTION CORE & SHELL	2025.03.05
REV PER TOOB COMMENTS	2024.11.22
ISSUED FOR PERMIT	2024.07.31
REVISION:	DATE:

SEAL:

STATE OF NEW YORK · RICHARD DENNIS REUTHER · 097739 · LICENSED PROFESSIONAL ENGINEER

DRAWING TITLE:

MECHANICAL PIPING DIAGRAMS

DATE: 2024.01.29

DRAWN BY: RK/BMF

PROJECT NO.: 24.03

DRAWING NO.: M5.1

TOB APPLICATION No. 22030355

Quick Results

Connection Summary

Indoor Units:	1 / 1 to 1
Capacity:	12 / 6 to 12 (100.0%)
* Connectable capacity is not actual capacity.	
Total Pipe Length:	25.9 / 100.0 feet

Refrigerant

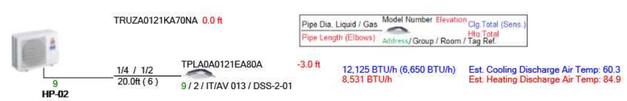
System Refrigerant:	**R410A**
Additional Field-Charge:	**0.00 lb**
Total System Charge:	**4.44 lb**

Correction Factors

	Cooling	Heating
Outdoor Temperature:	1.03	0.61
Piping Length:	0.99	1.00
Defrosting:	-	1.00
User Derate:	1.00	1.00
Outdoor Altitude:	1.00	1.00
	1.01	**0.61**

Conditions

Cooling		Heating	
Indoor Dry Bulb	72.0 °F	Indoor Dry Bulb	70.0 °F
Indoor Wet Bulb	67.0 °F		
Indoor Humidity	77.8 %	Outdoor Dry Bulb	12.0 °F
		Outdoor Wet Bulb	11.3 °F
Outdoor Dry Bulb	90.0 °F	Outdoor Humidity	85.2 %



1 M5.2 VRF PIPING DIAGRAM- HP-02

SCALE: NTS

Quick Results

Connection Summary

Indoor Units:	2 / 1 to 18	
Capacity:	72 / 36 to 108 (100.0%)	
* Connectable capacity is not actual capacity.		
Total Pipe Length:	299.9 / 1509.4	feet
Furthest Actual:	240.0 / 541.0	feet
Furthest Equiv.:	263.0 / 623.0	feet
Furthest IU from BC Actual:	90.0 / 197.0	feet
Furthest IU from BC Equiv.:	96.9 / 197.0	feet
Furthest IU from BC Thru Sub BC Actual:	0.0 / 0.0	feet
Furthest IU from BC Thru Sub BC Equiv.:	0.0 / 0.0	feet

Refrigerant

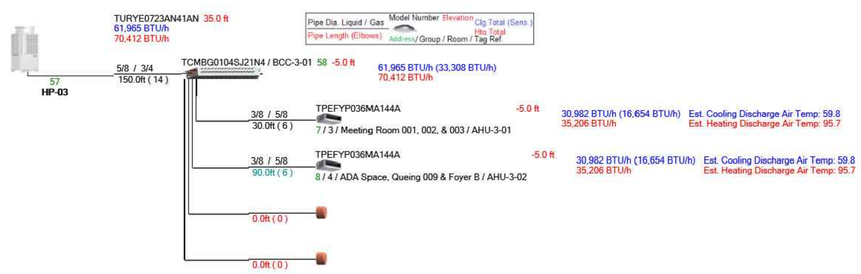
System Refrigerant:	**R410A**
Additional Field-Charge:	**26.98 lb**
Total System Charge:	**38.45 lb**

Correction Factors

	Cooling	Heating
Outdoor Unit Capacity:	1.00	1.00
Outdoor Temperature:	1.02	0.97
Piping Length:	0.84	0.96
Defrosting:	-	0.95
User Derate:	1.00	1.00
Outdoor Altitude:	1.00	1.00
	0.86	**0.88**

Conditions

Cooling		Heating	
Indoor Dry Bulb	72.0 °F	Indoor Dry Bulb	70.0 °F
Indoor Wet Bulb	67.0 °F		
Indoor Humidity	77.8 %	Outdoor Dry Bulb	12.0 °F
		Outdoor Wet Bulb	11.3 °F
Outdoor Dry Bulb	90.0 °F	Outdoor Humidity	85.2 %



2 M5.2 VRF PIPING DIAGRAM- HP-03

SCALE: NTS

PROJECT NAME / LOCATION:

PROPOSED "INTERIOR AND EXTERIOR ALTERATIONS"

India Pentecostal Assembly

310A SOUTH OYSTER BAY ROAD
SYOSSET, NY 11791

TOWN OF OYSTER BAY

ARCHITECT:

the KENT ARCHITECTURAL COLLABORATIVE, PLLC
132 CLYDE STREET, SUITE 1B
WEST SAYVILLE • NEW YORK • 11796
631.793.9684
rkent.tkac@gmail.com

CONSTRUCTION MANAGER / GENERAL CONTRACTOR:

CCC CORE CONCEPTS CONTRACTING
687 OLD WILLETS PATH, HAUPPAUGE, NY 11788
www.coreconceptscontracting.com 631.435.8985

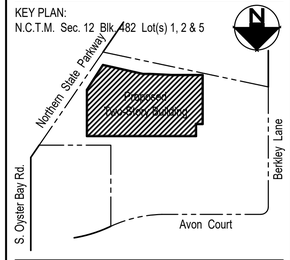
STRUCTURAL ENGINEER:

reuther&bowen Engineering, Design, Construction Services
326 WARD ST. SCRANTON, PA 18512-2424
PHONE (570) 496-7020 FAX (570) 496-7021

MECHANICAL / ELECTRICAL / PLUMBING ENGINEERS:

DMG designmanagementgroup building systems, smarter
2007 HIGHWAY 315, SUITE 202 PITTSTON, PA 18640
PHONE (570) 299-7520

KEY PLAN:
N.C.T.M. Sec. 12 Blk. 482 Lot(s) 1, 2 & 5



ISSUED FOR CONSTRUCTION CORE & SHELL	2025.03.05
REV PER TOOB COMMENTS	2024.11.22
ISSUED FOR PERMIT	2024.07.31
REVISION:	DATE:

SEAL:

DRAWING TITLE:

MECHANICAL PIPING DIAGRAMS

DATE: 2024.01.29

DRAWN BY: RK/BMF

PROJECT NO.: 24.03

DRAWING NO.: M5.2

TOB APPLICATION No. 22030355

Quick Results

Connection Summary

Indoor Units:	6 / 2 to 50	
Capacity:	288 / 144 to 432 (100.0%)	
* Connectable capacity is not actual capacity.		
Total Pipe Length:	799.2 / 3034.3	feet
Furthest Actual:	215.0 / 541.0	feet
Furthest Equiv.:	254.1 / 623.0	feet
Furthest IU from BC Actual:	150.0 / 197.0	feet
Furthest IU from BC Equiv.:	161.5 / 197.0	feet
Furthest IU from BC Thru Sub BC Actual:	0.0 / 0.0	feet
Furthest IU from BC Thru Sub BC Equiv.:	0.0 / 0.0	feet

Refrigerant

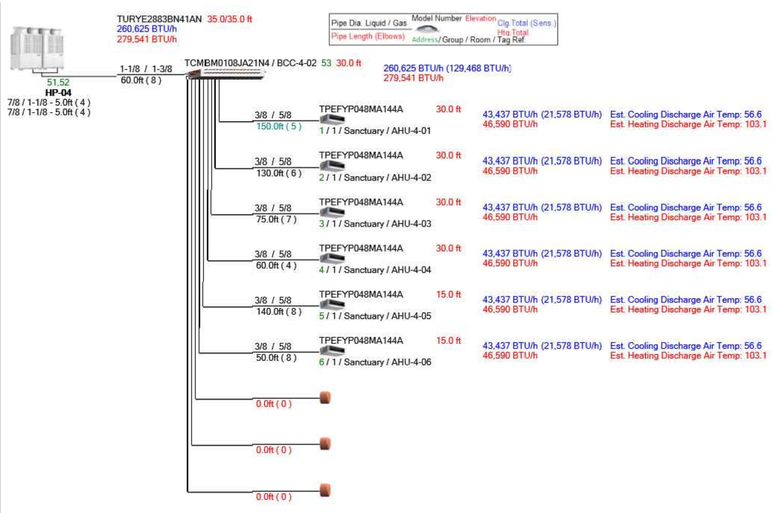
System Refrigerant:	**R410A**
Additional Field-Charge:	**61.34 lb**
Total System Charge:	**108.96 lb**

Correction Factors

	Cooling	Heating
Outdoor Unit Capacity:	1.00	1.00
Outdoor Temperature:	1.03	0.96
Piping Length:	0.88	0.95
Defrosting:	-	0.95
User Derate:	1.00	1.00
Outdoor Altitude:	1.00	1.00
	0.90	**0.87**

Conditions

Cooling		Heating	
Indoor Dry Bulb	72.0 °F	Indoor Dry Bulb	70.0 °F
Indoor Wet Bulb	67.0 °F		
Indoor Humidity	77.8 %	Outdoor Dry Bulb	12.0 °F
		Outdoor Wet Bulb	11.3 °F
Outdoor Dry Bulb	90.0 °F	Outdoor Humidity	85.2 %



1 / M5.3 VRF PIPING DIAGRAM- HP-04

SCALE: NTS

PROJECT NAME / LOCATION:

PROPOSED "INTERIOR AND EXTERIOR ALTERATIONS"

India Pentecostal Assembly

310A SOUTH OYSTER BAY ROAD
SYOSSET, NY 11791

TOWN OF OYSTER BAY

ARCHITECT:

the KENT ARCHITECTURAL COLLABORATIVE, PLLC
132 CLYDE STREET, SUITE 1B
WEST SAYVILLE • NEW YORK • 11796
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rkent.tkac@gmail.com

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STRUCTURAL ENGINEER:

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326 WARD ST. SCRANTON, PA 18512-2424
PHONE (570) 496-7020 FAX (570) 496-7021

MECHANICAL / ELECTRICAL / PLUMBING ENGINEERS:

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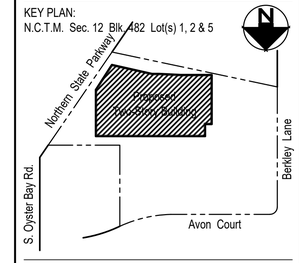
2007 HIGHWAY 315, SUITE 202 PITTSTON, PA 18640
PHONE (570) 299-7520

KEY PLAN:
N.C.T.M. Sec. 12 Blk. 482 Lot(s) 1, 2 & 5



ISSUED FOR CONSTRUCTION CORE & SHELL	2025.03.05
REV PER TOOB COMMENTS	2024.11.22
ISSUED FOR PERMIT	2024.07.31
REVISION:	DATE:

SEAL:

STATE OF NEW YORK · RICHARD DENNIS REUTHER · 097739 · LICENSED PROFESSIONAL ENGINEER

DRAWING TITLE:

MECHANICAL PIPING DIAGRAMS

DATE: 2024.01.29

DRAWN BY: RK/BMF

PROJECT NO.: 24.03

DRAWING NO.: M5.3

TOB APPLICATION No. 22030355

Quick Results

Connection Summary

Indoor Units:	6 / 1 to 48	
Capacity:	198 / 96 to 288 (103.1%)	
* Connectable capacity is not actual capacity.		
Total Pipe Length:	725.3 / 2460.6	feet
Furthest Actual:	180.0 / 541.0	feet
Furthest Equiv.:	206.2 / 623.0	feet
Furthest IU from BC Actual:	150.0 / 197.0	feet
Furthest IU from BC Equiv.:	163.1 / 197.0	feet
Furthest IU from BC Thru Sub BC Actual:	0.0 / 0.0	feet
Furthest IU from BC Thru Sub BC Equiv.:	0.0 / 0.0	feet

Refrigerant

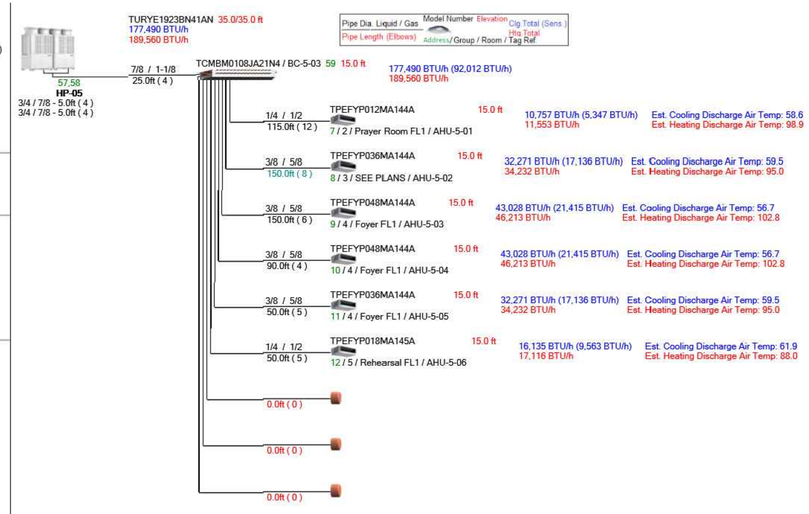
System Refrigerant:	**R410A**
Additional Field-Charge:	**44.50 lb**
Total System Charge:	**79.77 lb**

Correction Factors

	Cooling	Heating
Outdoor Unit Capacity:	1.01	1.00
Outdoor Temperature:	1.01	0.97
Piping Length:	0.90	0.96
Defrosting:	-	0.95
User Derate:	1.00	1.00
Outdoor Altitude:	1.00	1.00
	0.92	**0.88**

Conditions

Cooling		Heating	
Indoor Dry Bulb	72.0 °F	Indoor Dry Bulb	70.0 °F
Indoor Wet Bulb	67.0 °F		
Indoor Humidity	77.8 %	Outdoor Dry Bulb	12.0 °F
		Outdoor Wet Bulb	11.3 °F
Outdoor Dry Bulb	90.0 °F	Outdoor Humidity	85.2 %



1 / M5.4 VRF PIPING DIAGRAM- HP-05

SCALE: NTS

PROJECT NAME / LOCATION:

PROPOSED "INTERIOR AND EXTERIOR ALTERATIONS"

India Pentecostal Assembly

310A SOUTH OYSTER BAY ROAD
SYOSSET, NY 11791

TOWN OF OYSTER BAY

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STRUCTURAL ENGINEER:

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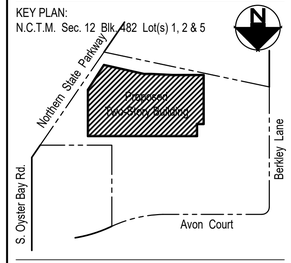
PHONE (570) 299-7520

KEY PLAN:
N.C.T.M. Sec. 12 Blk. 482 Lot(s) 1, 2 & 5



ISSUED FOR CONSTRUCTION CORE & SHELL	2025.03.05
REV PER TOOB COMMENTS	2024.11.22
ISSUED FOR PERMIT	2024.07.31
REVISION:	DATE:

SEAL:

STATE OF NEW YORK · RICHARD DENNIS REBERT · 097739 · LICENSED PROFESSIONAL ENGINEER

DRAWING TITLE:

MECHANICAL PIPING DIAGRAMS

DATE: 2024.01.29

DRAWN BY: RK/BMF

PROJECT NO.: 24.03

DRAWING NO.: M5.4

TOB APPLICATION No. 22030355

Quick Results

Connection Summary

Indoor Units:	8 / 2 to 50	
Capacity:	228 / 108 to 324 (105.6%)	
* Connectable capacity is not actual capacity.		
Total Pipe Length:	756.9 / 2624.7	feet
Furthest Actual:	160.0 / 541.0	feet
Furthest Equiv.:	189.9 / 623.0	feet
Furthest IU from BC Actual:	135.0 / 197.0	feet
Furthest IU from BC Equiv.:	148.8 / 197.0	feet
Furthest IU from BC Thru Sub BC Actual:	0.0 / 0.0	feet
Furthest IU from BC Thru Sub BC Equiv.:	0.0 / 0.0	feet

Refrigerant

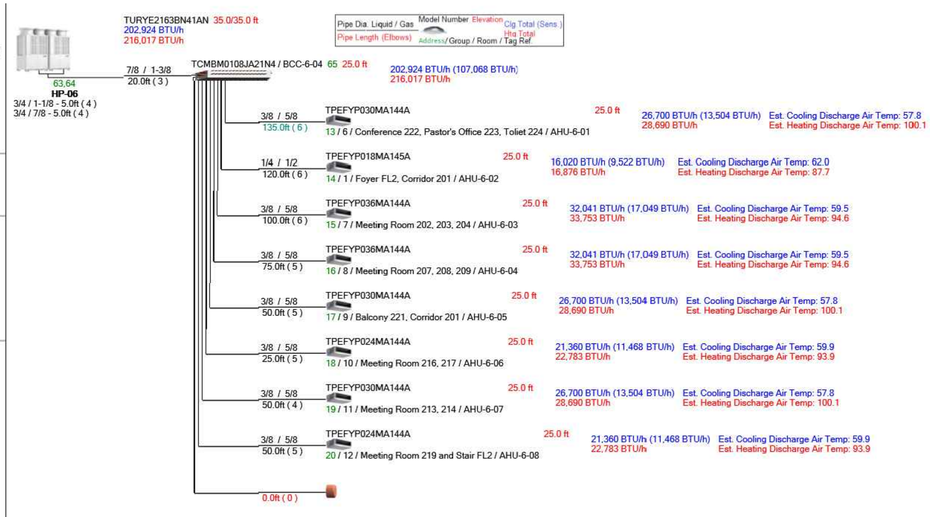
System Refrigerant:	**R410A**	
Additional Field-Charge:	**51.05**	**lb**
Total System Charge:	**86.33**	**lb**

Correction Factors

	Cooling	Heating
Outdoor Unit Capacity:	1.01	1.00
Outdoor Temperature:	1.01	0.96
Piping Length:	0.91	0.97
Defrosting:	-	0.95
User Derate:	1.00	1.00
Outdoor Altitude:	1.00	1.00
	0.94	**0.89**

Conditions

Cooling		Heating	
Indoor Dry Bulb	72.0 °F	Indoor Dry Bulb	70.0 °F
Indoor Wet Bulb	67.0 °F		
Indoor Humidity	77.8 %	Outdoor Dry Bulb	12.0 °F
		Outdoor Wet Bulb	11.3 °F
Outdoor Dry Bulb	90.0 °F	Outdoor Humidity	85.2 %



1 / M5.5 VRF PIPING DIAGRAM- HP-06

SCALE: NTS

PROJECT NAME / LOCATION:

PROPOSED "INTERIOR AND EXTERIOR ALTERATIONS"

India Pentecostal Assembly

310A SOUTH OYSTER BAY ROAD
SYOSSET, NY 11791

TOWN OF OYSTER BAY

ARCHITECT:

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MECHANICAL / ELECTRICAL / PLUMBING ENGINEERS:

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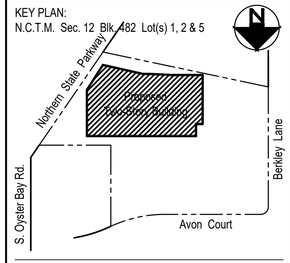
PHONE (570) 299-7520

KEY PLAN:
N.C.T.M. Sec. 12 Blk. 482 Lot(s) 1, 2 & 5



ISSUED FOR CONSTRUCTION CORE & SHELL	2025.03.05
REV PER TOOB COMMENTS	2024.11.22
ISSUED FOR PERMIT	2024.07.31
REVISION:	DATE:

SEAL:

DRAWING TITLE:

MECHANICAL PIPING DIAGRAMS

DATE: 2024.01.29

DRAWN BY: RK/BMF

PROJECT NO.: 24.03

DRAWING NO.: M5.5

TOB APPLICATION No. 22030355

DIAGRAM SYMBOL LEGEND		CONT.No		PAGE	
DISPLAY	DESCRIPTION				
—//—	POWER WIRE				
———	CONTROL WIRE				
— — —	REF. PIPE				

CITY MULTI

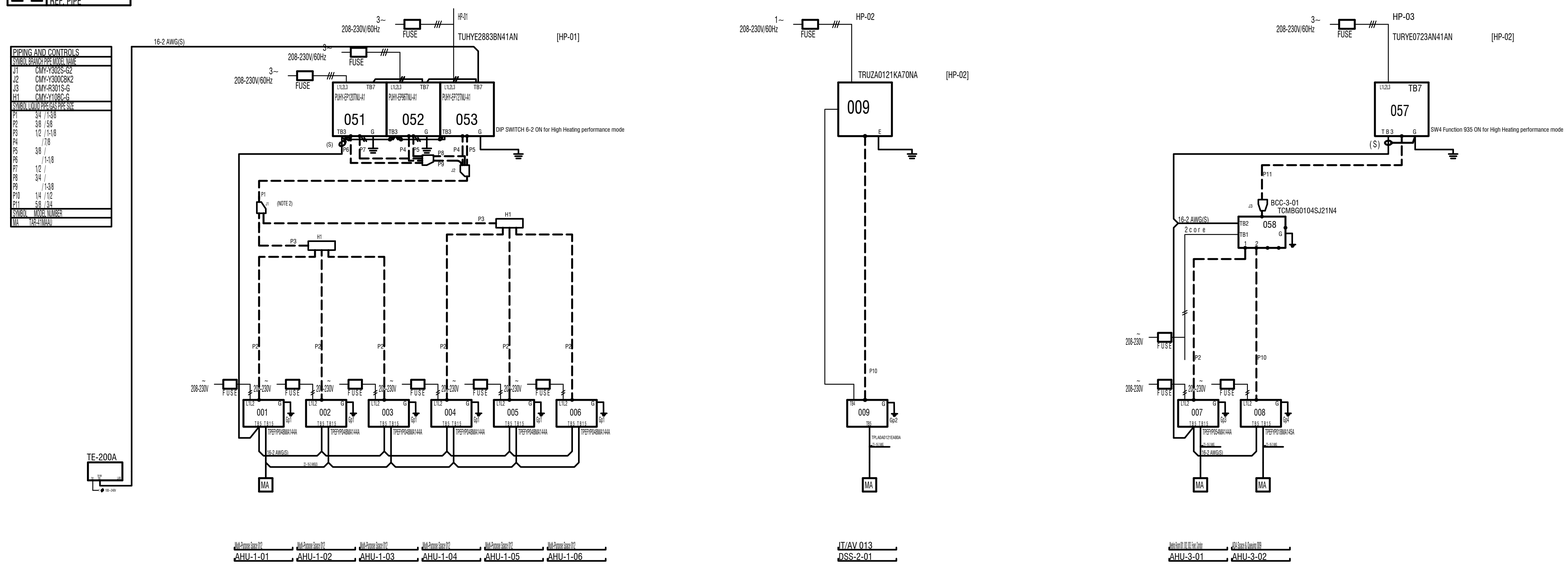
SYSTEM SCHEMATIC DWG.

This drawing is schematic in nature. Final routing of piping & wiring shall be determined by the installing contractor and/or designer of record

Additional refrigerant charge is needed depending on the size and length of extended piping. Please refer the amount of pre-charge and the formula of calculation which is mentioned on the data book.

1.25mm(16 AWG) : 1.25mm(16 AWG) or more. 0.75mm(20 AWG) : between 0.5mm(24 AWG) and 0.75mm(20 AWG).

Control Notes:
NOTE 1: Install twinning Y's within 15 degrees of level and with 20 inches of straight pipe on converging connection - reference installation manual for additional details including but not limited to special trapping requirements when twinning, and pipe slope requirements
NOTE 2: CMY-Y305S-G2 or CMY-Y302S-G2 pipe joints should be installed within 15 degrees of horizontal or vertical. When vertical the tee joint should not face downward - reference installation manual for additional details



1 M6.1 VRF PIPING DIAGRAM- HP-01,02,03

SCALE: NTS

REMARKS
Comments:

DIAGRAM SYMBOL LEGEND		CONT.No		PAGE	
DISPLAY	DESCRIPTION				
—//—	POWER WIRE				
———	CONTROL WIRE				
— — —	REF. PIPE				

CITY MULTI

SYSTEM SCHEMATIC DWG.

This drawing is schematic in nature. Final routing of piping & wiring shall be determined by the installing contractor and/or designer of record

Additional refrigerant charge is needed depending on the size and length of extended piping. Please refer the amount of pre-charge and the formula of calculation which is mentioned on the data book.

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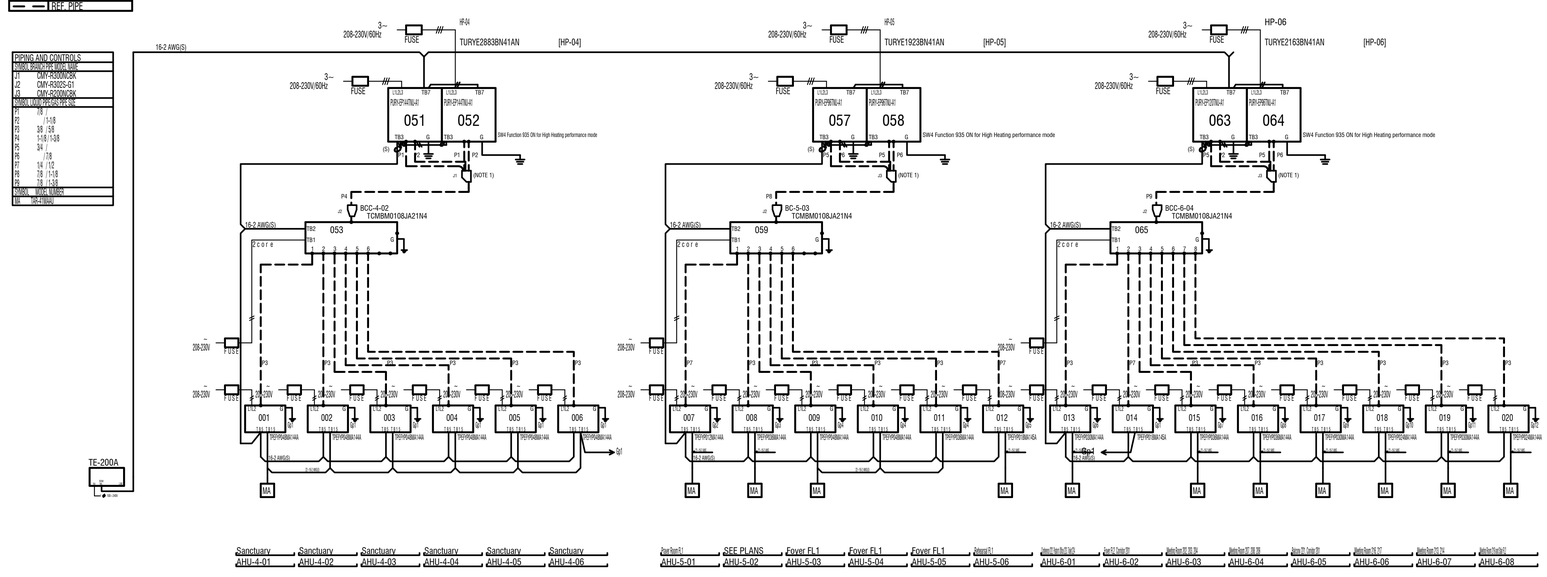
Control Notes:
NOTE 1: Install twinning Y's within 15 degrees of level and with 20 inches of straight pipe on converging connection - reference installation manual for additional details including but not limited to special trapping requirements when twinning, and pipe slope requirements



2 M6.1 VRF PIPING DIAGRAM- HP-04,05,06

SCALE: NTS

REMARKS
Comments:

PROJECT NAME / LOCATION:

PROPOSED "INTERIOR AND EXTERIOR ALTERATIONS"

India Pentecostal Assembly

310A SOUTH OYSTER BAY ROAD
SYOSSET, NY 11791

TOWN OF OYSTER BAY

ARCHITECT:

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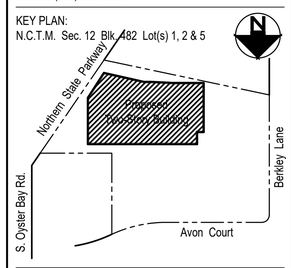
KEY PLAN:
N.C.T.M. Sec. 12 Blk. 482 Lot(s) 1, 2 & 5



ISSUED FOR CONSTRUCTION CORE & SHELL	2025.03.05
REV PER TOOB COMMENTS	2024.11.22
ISSUED FOR PERMIT	2024.07.31
REVISION:	DATE:

SEAL:

STATE OF NEW YORK · RICHARD DENNIS GEERITZ · 097739 · LICENSED PROFESSIONAL ENGINEER

DRAWING TITLE:
MECHANICAL WIRING DIAGRAMS

DATE: 2024.01.29

DRAWN BY: RK/BMF

PROJECT NO.: 24.03

DRAWING NO.: M6.1

TOB APPLICATION No. 22030355

MITSUBISHI ELECTRIC TRANE HVAC US: CITY MULTI VRF OUTDOOR UNIT SCHEDULE

System Tag	Tag Reference	M-NET Address	Model Number	Modules	Nominal Cooling Capacity (BTU/h)	Nominal Heating Capacity (BTU/h)	Cooling Efficiency (I)EER [SEER]	EER2 [SEER2]	Heating COP @ 47°F [HSPF]	Nom System Connected Capacity (% of NOM)	Design Cooling Outdoor Temp DB (°F)	Design Heating Outdoor Temp WB (°F)	Max Pipe Length from BC or 1st Joint (feet)	Refrig Pipe Dim High/Low Pressure (inch) (See Note 4)	Corrected Cooling Total Capacity (BTU/h)	Corrected Heating Capacity (BTU/h)	Sound Pressure (dBA)	Inverter Driven Compressor Type / Quantity	Preliminary Added Field Charge (See Note 5)	Electrical-Per Module 208/230 or [460V] Voltage / Phase	MCA 208/230 or [460V]	RFS	MOCP	Notes / Options
HP-01	HP-01	51, 52, 53	TUHYE2883BN41AN	EP120, EP96, EP72	288,000	320,000	22.05 / 10.75	0	3.780	100.0 %	90.0	11.3	78.4	3/4 / 1-3/8	279,433.7	288,407.2	82/83.5	SCROLL/3	33.6	208/230V / 3-phase 3-wire	55/49, 44/40, 32/29	60/50, 45/40, 35/30	90/80, 70/60, 50/45	1, 2, 3, 4, 5, 6, 7, 8, 9, 10
HP-02	HP-02	9	TRUZA0121KA70NA		12,000	14,000	16.4 []	0 [26.9]	4.94 [10.9]	100.0 %	90.0	11.3	0.0	1/4 / 1/2	12,124.6	8,531.0	44/46		0.0	208/230V / 1-phase	11	15	28	1, 2, 3, 4, 5, 6, 7, 8, 9
HP-03	HP-02	57	TURYE0723AN41AN	EP72	72,000	80,000	22.85 / 12	0	4.090	100.0 %	90.0	11.3	54.6	5/8 / 3/4	71,128.5	72,999.0	75.5/77.0	SCROLL/1	18.1	208/230V / 3-phase 3-wire	33/30	35/30	50/50	1, 2, 3, 4, 5, 6, 7, 8, 9
HP-04	HP-04	51, 52	TURYE2883BN41AN	EP144, EP144	288,000	320,000	20.1 / 9.35	0	3.360	100.0 %	90.0	11.3	123.0	1-1/8 / 1-3/8	274,806.8	285,357.5	88.5/88.5	SCROLL/2	46.4	208/230V / 3-phase 3-wire	60/60, 60/60	60/60, 60/60	100/100, 100/100	1, 2, 3, 4, 5, 6, 7, 8, 9
HP-05	HP-05	57, 58	TURYE1923BN41AN	EP96, EP96	192,000	215,000	21.45 / 11.15	0	3.825	103.1 %	90.0	11.3	56.6	7/8 / 1-1/8	189,774.1	194,486.4	80.5/82.0	SCROLL/2	42.7	208/230V / 3-phase 3-wire	44/40, 44/40	45/45, 45/45	70/60, 70/60	1, 2, 3, 4, 5, 6, 7, 8, 9
HP-06	HP-06	63, 64	TURYE2163BN41AN	EP120, EP96	216,000	243,000	20.9 / 10.45	0	3.725	105.6 %	90.0	11.3	59.2	7/8 / 1-3/8	214,052.6	219,923.9	82.0/83.0	SCROLL/2	45.3	208/230V / 3-phase 3-wire	56/55, 44/40	60/60, 45/45	90/90, 70/60	1, 2, 3, 4, 5, 6, 7, 8, 9

Notes & Options:

1. Nominal cooling capacities are based on indoor coil EAT of 80/67°F (DB/WB), outdoor of 95°F (DB)
2. Nominal heating capacities are based on indoor coil EAT of 70°F (DB), outdoor of 43°F (WB)
3. Efficiency values for EER, IEER, COP are based on AHRI 1230 test method for mixture of ducted & non-ducted indoor units.
4. For systems with multiple modules, refrigerant pipe dimensions indicate total system combined piping downstream of module twinning.
5. Added field charge listed is in addition to factory charge, this must be updated based upon final as-built piping layout.
6. Factory representatives shall review the project prior to and throughout the installation of CITY MULTI equipment
7. Factory representatives shall startup and commission CITY MULTI equipment upon completion of equipment installations
8. Factory representatives shall provide on-site assistance for the BMS integration of the CITY MULTI equipment
9. Factory representatives shall provide end-user training on the CITY MULTI equipment upon completion of the installation of equipment
10. Provide 18" Super Stand Kit

- Provide accessory Snow Hood, Hail Guards, and Panel Heaters
- Provide AE-200 Central Controller
- Snow Hood & Hail Guard for city multi recovery outdoor units (used for R2 units)
- Outdoor units shall be capable of providing 100% rated heating capacity at 18 F outdoor WB and 68% rated heat capacity at -13 F
- Y-Series heat pumps Provide ball valves for service downstream of all branches for R-410A, full bore, @ 600psia-800psia
- P-Series – provide control interface to communicate with VRF system central controller; Provide low ambient kit for cooling operation to 0 F

VRF HEAT RECOVERY BRANCH CIRCUIT CONTROLLER

System Tag	Tag Reference	M-NET Address	Model Number	Type (double / Main / Sub)	Number of Ports	Connected Capacity to BC	Voltage / Phase	Power Cooling 208V/230V (kW)	Power Heating 208V/230V (kW)	MCA 208/230	Notes / Options
HP-03	BCC-3-01	58	TCMBG0104SJ21N4	Single	4	72,000.0	208/230V/1-phase	0.061/0.078	0.3/0.039	0.38/0.44	1
HP-04	BCC-4-02	53	TCMBM0108JA21N4	Main	8	288,000.0	208/230V/1-phase	0.137/0.176	0.076/0.098	0.83/0.97	1
HP-05	BC-5-03	59	TCMBM0108JA21N4	Main	8	198,000.0	208/230V/1-phase	0.137/0.176	0.076/0.098	0.83/0.97	1
HP-06	BCC-6-04	65	TCMBM0108JA21N4	Main	8	228,000.0	208/230V/1-phase	0.137/0.176	0.076/0.098	0.83/0.97	1

Notes & Options:

1. Include Diamondback Ball Valves BV-Series, 700PSIG working pressure, full port, 410A rated.
2. For sub BC controller CMB-P-NU-GB1 or -GB, the total connectable indoor unit capacity can be 126,000 BTUs or less. If two sub BC controllers are used, the total indoor unit capacity connected to BOTH sub BC controllers also cannot exceed 126,000 BTUs. For sub BC controller CMB-P1016NU-HB1 the total connectable indoor unit capacity can be 126,000 BTUs or less. However, if two sub controllers are used, and one of them is CMB-1016NU-HB1, the total indoor unit capacity connected to BOTH sub controllers must NOT exceed 168,000 BTUs.
3. Provide Refrigeration Ball Valve-Braze/Schrader/Insulated - 3/8" size
4. Provide Refrigeration Ball Valve-Braze/Schrader/Insulated - 5/8" size

- Provide ball valves on all ports branches for R-410A, full bore, @ 600psia-800psia.
- Each BC controller shall include one (1) unused branch port for future use
- CONDENSATE DRAIN

DIFFUSER, GRILLE, AND REGISTER SCHEDULE

MARK	CFM	BASE OF DESIGN MANUFACTURER	BASE OF DESIGN MODEL	TYPE	THROW T150-T100-T50 (FT)	NECK SIZE (IN)	MODULE SIZE (IN)	MAX. NC LEVEL	MAX. SP LEVEL	NOTES
S-1	0-175	PRICE	SCD	CEILING SUPPLY DIFFUSER	4-7-10	6"Ø	12 SQ.	20	0.10	1,2,3,4
S-2	0-175	PRICE	SCD	CEILING SUPPLY DIFFUSER	3-4-7	6"Ø	24 SQ.	19	0.10	1,2,3,4
S-3	140-315	PRICE	SCD	CEILING SUPPLY DIFFUSER	3-6-10	8"Ø	24 SQ.	22	0.10	1,2,3,4
S-4	315-665	PRICE	SCD	CEILING SUPPLY DIFFUSER	5-8-13	12"Ø	24 SQ.	24	0.10	1,2,3,4
S-5	620-1000	PRICE	SDS-150	CEILING SUPPLY DIFFUSER	8-12-17	14"Ø	24 SQ.	30	0.10	1,2,3,4
S-6	330-500	PRICE	SDS-150 3 SLOT	LINEAR SUPPLY DIFFUSER	12-18-26	12"Ø	7-3/16" x 24 SQ.	30	0.10	1,2,3,4,5
S-7	505-700	PRICE	SDS-150 4 SLOT	LINEAR SUPPLY DIFFUSER	13-19-29	14"Ø	7-3/16" x 24 SQ.	30	0.10	1,2,3,4,5
R-1	0-500	PRICE	80	1/2"x1/2"x1/2" EGG CRATE	N/A	10x10	12x12	-	0.05	1,2,3
R-2	500-1000	PRICE	80	1/2"x1/2"x1/2" EGG CRATE	N/A	22x10	24x12	-	0.05	1,2,3
R-3	1000-2100	PRICE	81	1/2"x1/2"x1/2" EGG CRATE	N/A	22x22	24x24	-	0.05	1,2,3
E-1	0-500	PRICE	80	1/2"x1/2"x1/2" EGG CRATE	N/A	10x10	12x12	-	0.05	1,2,3
E-2	500-1000	PRICE	80	1/2"x1/2"x1/2" EGG CRATE	N/A	22x10	24x12	-	0.05	1,2,3
E-3	1000-2100	PRICE	81	1/2"x1/2"x1/2" EGG CRATE	N/A	22x22	24x24	-	0.05	1,2,3
E-4	0-1200	PRICE	80	1/2"x1/2"x1/2" EGG CRATE	N/A	22x12	24x14	-	0.07	1,2,3
T-1	0-150	PRICE	ATG1	TRANSFER GRILLE	N/A	12x10	12x10	-	0.05	1,3
T-2	150-300	PRICE	ATG1	TRANSFER GRILLE	N/A	24x10	24x10	-	0.05	1,3
T-3	300-700	PRICE	ATG1	TRANSFER GRILLE	N/A	24x22	24x22	-	0.05	1,3

NOTES:

1. INSTALL IN ACCORDANCE WITH MANUFACTURER WRITTEN INSTRUCTIONS. COORDINATE WITH ALL TRADES.
2. PROVIDE WITH FACTORY INSTALLED OPPOSED BLADE DAMPER.
3. COORDINATE MOUNTING HARDWARE WITH ARCHITECTURAL CEILING AND WALL FINISHES
4. DUCT RUNNOUT SIZES SHALL MATCH DIFFUSER OR GRILLE CONNECTION DIMENSIONS UNLESS OTHERWISE NOTED.
5. PROVIDE WITH PLENUM BOX.

ELECTRICAL WALL HEATER SCHEDULE

TAG	BASIS OF DESIGN MANUFACTURER	BASIS OF DESIGN MODEL	AREA SERVED	CFM	BTU/HR	KW	NO. OF FANS	ELECTRICAL VOLT/PH/HZ	ELECTRICAL MCA	ELECTRICAL MOCP	UNIT SIZE (IN.) W x H x D	NOTES
EWH-01	MARKEL	F3032	SEE PLANS	100	3413	1.0	1	208/1/60	4.8	15	9-1/4" X 12-1/4" X 3-3/4"	1,2,3
EWH-02	MARKEL	H3032	SEE PLANS	100	5120	1.5	1	208/1/60	7.3	15	9-1/4" X 12-1/4" X 3-3/4"	1,2,3

NOTES:

1. INSTALL IN ACCORDANCE WITH MANUFACTURER'S INSTRUCTIONS. COORDINATE INSTALLATION WITH ALL OTHER TRADES.
2. PROVIDE COMPLETE WITH BUILT-IN DISCONNECT SWITCH, BUILT-IN THERMOSTAT CUTOUTS, WIRING, AND ACCESSORIES.
3. COLOR SELECTED BY ARCHITECT FROM MANUFACTURER'S STANDARD COLORS.

[FUTURE] KITCHEN EXHAUST FAN SCHEDULE

TAG	AREA SERVED	BASE OF DESIGN MANUFACTURER	BASE OF DESIGN MODEL	FAN TYPE	DRIVE	CFM	ESP	RPM	HP	BHP	ELECTRICAL VOLT/PH/HZ	ELECTRICAL FLA	ELECTRICAL MCA	ELECTRICAL MOCP	ROOF OPENING (IN.)	WEIGHT (LBS.)	NOTES
KEF-1	KITCHEN	GREENHECK	CUE-240-VGD	UPBLAST ROOF	DIRECT	5,600	1.25	860	3.0	2.10	208/3/60	8.0	10.0	20	30.5x30.5	253	1,2,3
KEF-2	KITCHEN	GREENHECK	CUE-100HP-VG	UPBLAST ROOF	DIRECT	300	1.00	1,706	1/4	0.13	115/1/60	2.9	4.0	15	15.5x15.5	66	1,2,3

NOTES:

1. INSTALL IN ACCORDANCE WITH MANUFACTURER WRITTEN INSTRUCTIONS. COORDINATE INSTALLATION WITH ALL TRADES.
2. PROVIDE WITH 12" ROOF CURB, GREASE DRAIN TROUGH, CLEAN-OUT PORT, HEAT BAFFLE, NON-STICK WHEEL, WALL BRACKET, VFD RATED MOTOR, AND TOGGLE SWITCH.
3. INTERLOCK FAN WITH MAKEUP AIR UNIT KMAU-1; MONITOR BY BMS.

[FUTURE] KITCHEN MAKEUP AIR UNIT (DX & GAS) SCHEDULE

TAG	AREA SERVED	BASIS OF DESIGN MANUFACTURER	BASIS OF DESIGN MODEL	TOTAL CFM	SUPPLY FAN CFM	MOTOR QTY.	HP (EA.)	BHP	ESP (IN. W.C.)	RPM	COOLING COIL TOTAL (MBH)	SENS. (MBH)	EAT (DB/WB)	LAT (DB/WB)	EER	DIRECT HEATING COIL FUEL TYPE	INPUT (MBH)	OUTPUT (MBH)	EFF. (%)	EAT (DB)	LAT (DB)	STAGES	CONN. SIZE	ELECTRICAL VOLTS/PH/Hz	MCA	MOCP	WEIGHT (LBS.)	NOTES
KMAU-1	KITCHEN	GREENHECK	DGX-P122-H22-MF-7	5,050	5,050	1	5	3.25	0.75	1,287	120.4	78.2	88.9/75.0	74.8/69.0	17.0	NAT. GAS	394.2	362.7	92.0	4.0	70.0	MOD.	3/4"	208/3/60	54.8	70	2,737	1,2,3

NOTES:

1. PROVIDE COMPLETE PACKAGED UNIT WITH ALL CONTROLS AND ACCESSORIES FOR PROPER OPERATION, INCLUDING 6" EQUIPMENT PAD, DISCHARGE TEMPERATURE CONTROL, VFD SUPPLY FAN, INLET AIR SENSORS, FREEZE PROTECTION, MERV-8 FILTRATION, LOW LEAKAGE DAMPER, AND POWERED GFI CONVENIENCE OUTLET.
2. PROVIDE UL OR ETL LISTED CONTROL PANEL AND FUSED DISCONNECT FOR SINGLE POINT POWER CONNECTION.
3. PROVIDE UNIT COLOR AS SELECTED BY ARCHITECT FROM MANUFACTURER'S STANDARD COLORS.

ENERGY RECOVERY UNIT (DX & GAS) SCHEDULE

TAG	AREA SERVED	BASIS OF DESIGN MANUFACTURER	BASIS OF DESIGN MODEL	TOTAL CFM	SUPPLY FAN CFM	MOTOR QTY.	HP (EA.)	BHP	ESP (IN. W.C.)	RPM	EXHAUST FAN CFM	MOTOR QTY.	HP (EA.)	BHP	ESP	RPM	COOLING COIL TOTAL (MBH)	SENS. (MBH)	EAT (DB/WB)	LAT (DB/WB)	HOT GAS REHEAT CAPACITY (MBH)	LAT (DB)	INDIRECT HEATING COIL FUEL TYPE	INPUT (MBH)	OUTPUT (MBH)	EFF. (%)	EAT (DB)	LAT (DB)	STAGES	CONN. SIZE	COMPRESSORS NO.	RLA	LRA	TYPE	WHEEL/WINTER EAT (DB/WB)	WHEEL/WINTER LAT (DB/WB)	WHEEL/SUMMER EAT (DB/WB)	WHEEL/SUMMER LAT (DB/WB)	ELECTRICAL VOLTS/PH/Hz	MCA	MOCP	WEIGHT (LBS.)	NOTES
ERU-1	BASEMENT	GREENHECK	RVE-85-52D-20I-J-A1	5,375	5,375	1	7.5	5.32	1.50	1,717	4,800	1	5.0	3.04	1.25	1,418	286.6	172.5	80.1/67.9	50.9/50.9	191.3	83.9	NAT. GAS	300.0	243.0	81.0	43.7	85.5	4:1	3/4"	2	32.3/34.0	N/A/240	INVERTER SCROLL	3.5/1.7	43.7/37.0	88.9/75.0	80.1/68.0	208/3/60	123.4	150	4,577	1,2,3,4
ERU-2	BASEMENT	GREENHECK	RVE-85-52D-20I-J-A1	5,375	5,375	1	7.5	5.32	1.50	1,717	4,800	1	5.0	3.04	1.25	1,418	286.6	172.5	80.1/67.9	50.9/50.9	191.3	83.9	NAT. GAS	300.0	243.0	81.0	43.7	85.5	4:1	3/4"	2	32.3/34.0	N/A/240	INVERTER SCROLL	3.5/1.7	43.7/37.0	88.9/75.0	80.1/68.0	208/3/60	123.4	150	4,577	1,2,3,4
ERU-3	FIRST / SECOND FLOORS	GREENHECK	RVE-85-52D-20I-J-A1	6,370	6,370	1	10.0	7.45	1.50	1,770	5,775	1	5.0	4.20	1.25	1,760	298.8	186.4	80.5/68.4	53.9/53.8	213.2	84.9	NAT. GAS	300.0	243.0	81.0	41.7	77.0	4:1	3/4"	2	32.3/34.0	N/A/240	INVERTER SCROLL	3.5/1.7	42.9/36.4	88.9/75.0	80.5/68.4	208/3/60	129.2	150	4,528	1,2,3,4

NOTES:

1. PROVIDE COMPLETE PACKAGED UNIT WITH ALL CONTROLS AND ACCESSORIES FOR PROPER OPERATION, INCLUDING 14" ROOF CURB, BUILDING STATIC PRESSURE SENSOR, SPACE HUMIDITY SENSOR, OUTSIDE AIR TEMPERATURE SENSOR, SUPPLY AIR TEMPERATURE SENSOR, SPACE TEMPERATURE SENSOR, HINGED ACCESS PANELS, VFD RATED DIRECT DRIVE MOTORS, CO2 SENSOR AND SUPPLY FAN CONTROL, DDC ACTUATOR, ENERGY WHEEL ECONOMIZER CONTROL, FROST CONTROL MODULATING WHEEL, LOW LEAKAGE DAMPERS, MERV-13 FILTRATION, REMOTE DISPLAY (150FT CORD), AND POWERED GFI CONVENIENCE OUTLET.
2. PROVIDE UL OR ETL LISTED WITH CONTROL PANEL AND FUSED DISCONNECT FOR SINGLE POINT POWER CONNECTION.
3. INSTALL SUPPLY AND RETURN AIR SMOKE DETECTOR FOR ALL UNITS WITH GREATER THAN 2000 CFM IN ACCORDANCE WITH NFPA 72 AND MANUFACTURER'S INSTRUCTIONS. SMOKE DETECTOR SHALL BE FURNISHED AND WIRED BY E.C. (ELECTRICAL CONTRACTOR)
4. PROVIDE UNIT COLOR AS SELECTED BY ARCHITECT FROM MANUFACTURER'S STANDARD COLORS.
5. TRANE OR APPROVED EQUALS ARE ACCEPTABLE.

PROJECT NAME / LOCATION:

PROPOSED "INTERIOR AND EXTERIOR ALTERATIONS"
India Pentecostal Assembly
310A SOUTH OYSTER BAY ROAD
SYOSSET, NY 11791
TOWN OF OYSTER BAY

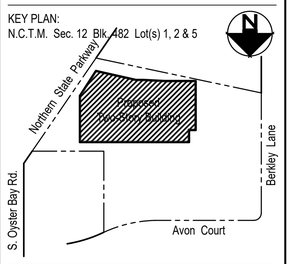
ARCHITECT:
the KENT ARCHITECTURAL COLLABORATIVE, PLLC
132 CLYDE STREET, SUITE 1B
WEST SAYVILLE • NEW YORK • 11796
631.793.9684
rkent.tkac@gmail.com

CONSTRUCTION MANAGER / GENERAL CONTRACTOR:
CORE CONCEPTS CONTRACTING
687 OLD WILLETS PATH, HAUPPAUGE, NY 11788
www.coreconceptscontracting.com 631.435.8985

STRUCTURAL ENGINEER:
reuther&bowen Engineering, Design, Construction Services
326 WARD ST. SCRANTON, PA 18512-2424
PHONE (570) 496-7020 FAX (570) 496-7021

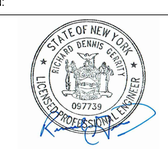
MECHANICAL / ELECTRICAL / PLUMBING ENGINEERS:
DMG designmanagementgroup building systems, smarter
2007 HIGHWAY 315, SUITE 202 PITTSTON, PA 18640
PHONE (570) 299-7520

KEY PLAN:
N.C.T.M. Sec. 12 Blk. 482 Lot(s) 1, 2 & 5



ISSUED FOR CONSTRUCTION CORE & SHELL	2025.03.05
REV PER TOOB COMMENTS	2024.11.22
ISSUED FOR PERMIT	2024.07.31
REVISION:	DATE:

SEAL:



DRAWING TITLE:
MECHANICAL SCHEDULES

DATE: 2024.01.29
DRAWN BY: RK/BMF
PROJECT NO.: 24.03
DRAWING NO.: M7.1

TOB APPLICATION No. 22030355

MITSUBISHI ELECTRIC TRANE HVAC US: CITY MULTI VRF INDOOR UNIT SCHEDULE

System Tag	Room Name	Tag Reference	Model	Type	Nominal Cooling Capacity (BTU/h)	Nominal Heating Capacity (BTU/h)	Cooling Design Entering Temp DB/WB (°F) / [Water in temp]	Heating Design Entering Temp DB/WB (°F) / [Water in temp]	Corrected Capacity: Cooling Diversity Full/Partial (See Note 5, 6)	Corrected Capacity: Cooling Total Capacity (BTU/h)	Corrected Capacity: Cooling Sensible Capacity (BTU/h)	Corrected Capacity: Heating Diversity Full/Partial (See Note 5, 6)	Corrected Capacity: Heating Capacity (BTU/h)	Estimated Cooling Coil LAT (°F) / [LWT]	Estimated Heating Coil LAT (°F) / [LWT]	Refrig Pipe Dim Liquid/Suction (inch)	Fan Speed Setting	Peak Fan Airflow (cfm) / [Design gpm G(US)/min]	Max Fan ESP Setting 208V/230V (IN WG)	Voltage / Phase	Power Cooling 208V/230V (kW)	Power Heating 208V/230V (kW)	Electrical MCA/MFS	Condensate Removal Rate (gal/hr)	Notes / Options
HP-01	Multi-Purpose Space 012	AHU-1-01	TPEFYP048MA144A	Ceiling-Concealed (Ducted)	48,000	54,000	72.0/67.0	70	FULL DEMAND	46,572.3	22,835.9	FULL DEMAND	48,067.9	55.7	104.1	3/8 / 5/8	HIGH	1306	0.6/0.6	208/230V/1-phase	0.242	0.24	4.38/15	1.53	1, 2, 3, 4
HP-01	Multi-Purpose Space 012	AHU-1-02	TPEFYP048MA144A	Ceiling-Concealed (Ducted)	48,000	54,000	72.0/67.0	70	FULL DEMAND	46,572.3	22,835.9	FULL DEMAND	48,067.9	55.7	104.1	3/8 / 5/8	HIGH	1306	0.6/0.6	208/230V/1-phase	0.242	0.24	4.38/15	1.53	1, 2, 3, 4
HP-01	Multi-Purpose Space 012	AHU-1-03	TPEFYP048MA144A	Ceiling-Concealed (Ducted)	48,000	54,000	72.0/67.0	70	FULL DEMAND	46,572.3	22,835.9	FULL DEMAND	48,067.9	55.7	104.1	3/8 / 5/8	HIGH	1306	0.6/0.6	208/230V/1-phase	0.242	0.24	4.38/15	1.53	1, 2, 3, 4
HP-01	Multi-Purpose Space 012	AHU-1-04	TPEFYP048MA144A	Ceiling-Concealed (Ducted)	48,000	54,000	72.0/67.0	70	FULL DEMAND	46,572.3	22,835.9	FULL DEMAND	48,067.9	55.7	104.1	3/8 / 5/8	HIGH	1306	0.6/0.6	208/230V/1-phase	0.242	0.24	4.38/15	1.53	1, 2, 3, 4
HP-01	Multi-Purpose Space 012	AHU-1-05	TPEFYP048MA144A	Ceiling-Concealed (Ducted)	48,000	54,000	72.0/67.0	70	FULL DEMAND	46,572.3	22,835.9	FULL DEMAND	48,067.9	55.7	104.1	3/8 / 5/8	HIGH	1306	0.6/0.6	208/230V/1-phase	0.242	0.24	4.38/15	1.53	1, 2, 3, 4
HP-01	Multi-Purpose Space 012	AHU-1-06	TPEFYP048MA144A	Ceiling-Concealed (Ducted)	48,000	54,000	72.0/67.0	70	FULL DEMAND	46,572.3	22,835.9	FULL DEMAND	48,067.9	55.7	104.1	3/8 / 5/8	HIGH	1306	0.6/0.6	208/230V/1-phase	0.242	0.24	4.38/15	1.53	1, 2, 3, 4
HP-02	IT/AV 013	DSS-2-01	TPLA0A0121EA80A	Ceiling-Cassette (Four-Way)	12,000	14,000	72.0/67.0	70	FULL DEMAND	12,124.6	6,650.2	FULL DEMAND	8,531.0	60.3	84.9	1/2 / 1/4	HIGH	530		208/230V/1-phase			Powered by Outdoor	0.16	1, 2, 3, 4
HP-03	Meeting Room 001, 002, & 003	AHU-3-01	TPEFYP036MA144A	Ceiling-Concealed (Ducted)	36,000	40,000	72.0/67.0	70	FULL DEMAND	35,564.3	18,383.2	FULL DEMAND	36,499.5	58.5	96.6	3/8 / 5/8	HIGH	1271	0.6/0.6	208/230V/1-phase	0.222	0.22	4.25/15	0.92	1, 2, 3, 4
HP-03	ADA Space, Queing 009 & Foyer B	AHU-3-02	TPEFYP036MA144A	Ceiling-Concealed (Ducted)	36,000	40,000	72.0/67.0	70	FULL DEMAND	35,564.3	18,383.2	FULL DEMAND	36,499.5	58.5	96.6	3/8 / 5/8	HIGH	1271	0.6/0.6	208/230V/1-phase	0.222	0.22	4.25/15	0.92	1, 2, 3, 4
HP-04	Sanctuary	AHU-4-01	TPEFYP048MA144A	Ceiling-Concealed (Ducted)	48,000	54,000	72.0/67.0	70	FULL DEMAND	45,801.1	22,524.3	FULL DEMAND	47,559.6	56.0	103.8	3/8 / 5/8	HIGH	1306	0.6/0.6	208/230V/1-phase	0.242	0.24	4.38/15	1.53	1, 2, 3, 4
HP-04	Sanctuary	AHU-4-02	TPEFYP048MA144A	Ceiling-Concealed (Ducted)	48,000	54,000	72.0/67.0	70	FULL DEMAND	45,801.1	22,524.3	FULL DEMAND	47,559.6	56.0	103.8	3/8 / 5/8	HIGH	1306	0.6/0.6	208/230V/1-phase	0.242	0.24	4.38/15	1.53	1, 2, 3, 4
HP-04	Sanctuary	AHU-4-03	TPEFYP048MA144A	Ceiling-Concealed (Ducted)	48,000	54,000	72.0/67.0	70	FULL DEMAND	45,801.1	22,524.3	FULL DEMAND	47,559.6	56.0	103.8	3/8 / 5/8	HIGH	1306	0.6/0.6	208/230V/1-phase	0.242	0.24	4.38/15	1.53	1, 2, 3, 4
HP-04	Sanctuary	AHU-4-04	TPEFYP048MA144A	Ceiling-Concealed (Ducted)	48,000	54,000	72.0/67.0	70	FULL DEMAND	45,801.1	22,524.3	FULL DEMAND	47,559.6	56.0	103.8	3/8 / 5/8	HIGH	1306	0.6/0.6	208/230V/1-phase	0.242	0.24	4.38/15	1.53	1, 2, 3, 4
HP-04	Sanctuary	AHU-4-05	TPEFYP048MA144A	Ceiling-Concealed (Ducted)	48,000	54,000	72.0/67.0	70	FULL DEMAND	45,801.1	22,524.3	FULL DEMAND	47,559.6	56.0	103.8	3/8 / 5/8	HIGH	1306	0.6/0.6	208/230V/1-phase	0.242	0.24	4.38/15	1.53	1, 2, 3, 4
HP-04	Sanctuary	AHU-4-06	TPEFYP048MA144A	Ceiling-Concealed (Ducted)	48,000	54,000	72.0/67.0	70	FULL DEMAND	45,801.1	22,524.3	FULL DEMAND	47,559.6	56.0	103.8	3/8 / 5/8	HIGH	1306	0.6/0.6	208/230V/1-phase	0.242	0.24	4.38/15	1.53	1, 2, 3, 4
HP-05	Prayer Room FL1	AHU-5-01	TPEFYP012MA144A	Ceiling-Concealed (Ducted)	12,000	13,500	72.0/67.0	70	FULL DEMAND	11,501.5	5,646.1	FULL DEMAND	11,853.6	57.8	99.6	1/4 / 1/2	HIGH	371	0.6/0.6	208/230V/1-phase	0.052	0.05	2.13/15	0.4	1, 2, 3, 4
HP-05	SEE PLANS	AHU-5-02	TPEFYP036MA144A	Ceiling-Concealed (Ducted)	36,000	40,000	72.0/67.0	70	FULL DEMAND	34,504.4	17,979.1	FULL DEMAND	35,121.7	58.8	95.6	3/8 / 5/8	HIGH	1271	0.6/0.6	208/230V/1-phase	0.222	0.22	4.25/15	0.92	1, 2, 3, 4
HP-05	Foyer FL1	AHU-5-03	TPEFYP048MA144A	Ceiling-Concealed (Ducted)	48,000	54,000	72.0/67.0	70	FULL DEMAND	46,005.8	22,606.9	FULL DEMAND	47,414.3	55.9	103.7	3/8 / 5/8	HIGH	1306	0.6/0.6	208/230V/1-phase	0.242	0.24	4.38/15	1.53	1, 2, 3, 4
HP-05	Foyer FL1	AHU-5-04	TPEFYP048MA144A	Ceiling-Concealed (Ducted)	48,000	54,000	72.0/67.0	70	FULL DEMAND	46,005.8	22,606.9	FULL DEMAND	47,414.3	55.9	103.7	3/8 / 5/8	HIGH	1306	0.6/0.6	208/230V/1-phase	0.242	0.24	4.38/15	1.53	1, 2, 3, 4
HP-05	Foyer FL1	AHU-5-05	TPEFYP036MA144A	Ceiling-Concealed (Ducted)	36,000	40,000	72.0/67.0	70	FULL DEMAND	34,504.4	17,979.1	FULL DEMAND	35,121.7	58.8	95.6	3/8 / 5/8	HIGH	1271	0.6/0.6	208/230V/1-phase	0.222	0.22	4.25/15	0.92	1, 2, 3, 4
HP-05	Rehearsal FL1	AHU-5-06	TPEFYP018MA145A	Ceiling-Concealed (Ducted)	18,000	20,000	72.0/67.0	70	FULL DEMAND	17,252.2	9,963.1	FULL DEMAND	17,560.9	61.5	88.4	1/4 / 1/2	HIGH	883	0.6/0.6	208/230V/1-phase	0.082	0.14	2.88/15	0.11	1, 2, 3, 4
HP-06	Conference 222, Pastor's Office 223, Toilet 224	AHU-6-01	TPEFYP030MA144A	Ceiling-Concealed (Ducted)	30,000	34,000	72.0/67.0	70	FULL DEMAND	28,164.8	14,078.4	FULL DEMAND	29,208.6	57.2	100.7	3/8 / 5/8	HIGH	883	0.6/0.6	208/230V/1-phase	0.142	0.14	2.88/15	1.04	1, 2, 3, 4
HP-06	Foyer FL2, Corridor 201	AHU-6-02	TPEFYP018MA145A	Ceiling-Concealed (Ducted)	18,000	20,000	72.0/67.0	70	FULL DEMAND	16,898.9	9,836.0	FULL DEMAND	17,181.6	61.6	88.0	1/4 / 1/2	HIGH	883	0.6/0.6	208/230V/1-phase	0.082	0.14	2.88/15	0.11	1, 2, 3, 4
HP-06	Meeting Room 202, 203, 204	AHU-6-03	TPEFYP036MA144A	Ceiling-Concealed (Ducted)	36,000	40,000	72.0/67.0	70	FULL DEMAND	33,797.8	17,711.1	FULL DEMAND	34,363.1	59.0	95.1	3/8 / 5/8	HIGH	1271	0.6/0.6	208/230V/1-phase	0.222	0.22	4.25/15	0.92	1, 2, 3, 4
HP-06	Meeting Room 207, 208, 209	AHU-6-04	TPEFYP036MA144A	Ceiling-Concealed (Ducted)	36,000	40,000	72.0/67.0	70	FULL DEMAND	33,797.8	17,711.1	FULL DEMAND	34,363.1	59.0	95.1	3/8 / 5/8	HIGH	1271	0.6/0.6	208/230V/1-phase	0.222	0.22	4.25/15	0.92	1, 2, 3, 4
HP-06	Balcony 221, Corridor 201	AHU-6-05	TPEFYP030MA144A	Ceiling-Concealed (Ducted)	30,000	34,000	72.0/67.0	70	FULL DEMAND	28,164.8	14,078.4	FULL DEMAND	29,208.6	57.2	100.7	3/8 / 5/8	HIGH	883	0.6/0.6	208/230V/1-phase	0.142	0.14	2.88/15	1.04	1, 2, 3, 4
HP-06	Meeting Room 216, 217	AHU-6-06	TPEFYP024MA144A	Ceiling-Concealed (Ducted)	24,000	27,000	72.0/67.0	70	FULL DEMAND	22,531.9	11,906.9	FULL DEMAND	23,195.1	59.5	94.4	3/8 / 5/8	HIGH	883	0.6/0.6	208/230V/1-phase	0.142	0.14	2.88/15	0.82	1, 2, 3, 4
HP-06	Meeting Room 213, 214	AHU-6-07	TPEFYP030MA144A	Ceiling-Concealed (Ducted)	30,000	34,000	72.0/67.0	70	FULL DEMAND	28,164.8	14,078.4	FULL DEMAND	29,208.6	57.2	100.7	3/8 / 5/8	HIGH	883	0.6/0.6	208/230V/1-phase	0.142	0.14	2.88/15	1.04	1, 2, 3, 4
HP-06	Meeting Room 219 and Stair FL2	AHU-6-08	TPEFYP024MA144A	Ceiling-Concealed (Ducted)	24,000	27,000	72.0/67.0	70	FULL DEMAND	22,531.9	11,906.9	FULL DEMAND	23,195.1	59.5	94.4	3/8 / 5/8	HIGH	883	0.6/0.6	208/230V/1-phase	0.142	0.14	2.88/15	0.82	1, 2, 3, 4

Notes & Options:

1 Nominal cooling capacities are based on indoor coil EAT of 80/67°F (DB/WB), outdoor of 95°F (DB)
2 Nominal heating capacities are based on indoor coil EAT of 70°F (DB), outdoor of 43°F (WB)
3 See outdoor unit schedule for outdoor ambient conditions, connected capacity, and other factors associated with corrected capacities
4 See schematic piping/control diagram for indication of required indoor unit remote controllers, system controllers, and integration devices.
5 Full demand corrected capacity includes de-rate associated with indoor vs. outdoor connected capacity indicated on outdoor unit schedule for associated system.
Partial corrected capacity assumes sufficient diversity exists such that the connected capacity de-rate does not apply.
It is the designer's responsibility to ensure "Diamond System Builder" is set in the appropriate output capacity setting (full demand/partial demand) prior to generating this schedule.
6 It is recommended to always base heating corrected capacity on full demand.

- Provide simple MA wired remote controller
- The indoor cassette & ducted concealed unit has a built in condensate lift mechanism so an external pump may not be necessary.
- Condensate pumps with safety switch alarm Field installed accessory for wall mount units and are powered by the indoor units.tion to 0 F

LOUVER SCHEDULE

TAG	BASIS OF DESIGN: MANUFACTURER	BASIS OF DESIGN: MODEL	AREA SERVED	CFM	TYPE	LENGTH	HEIGHT	FREE AREA	MAX FACE VELOCITY	NOTES
L-01	GREENHECK	ESD-403	109 PRAYER ROOM	75	INTAKE	12"	12"	0.20	378	1,2,3,4
L-02	GREENHECK	ESD-403	110 TOILET	75	INTAKE	12"	12"	0.20	378	1,2,3,4

NOTES:

1. PROVIDE WITH BACKDRAFT DAMPER
2. FIELD VERIFY EXACT SIZES OF ALL NEW LOUVERS IN EXISTING OPENINGS, APPROX. SIZE SHOWN.
3. LOUVERS SHALL BE FLANGELESS WITH SEALANT SLOT AROUND JAMB, HEAD, AND SILL WHEN LOUVER ATTACHES TO A WINDOW MULLION. PROVIDE SLEEVE (OF SAME MATERIAL AS LOUVER) TO ATTACH LOUVER TO MULLION.
4. PROVIDE UNIT WITH STANDARD COLOR AS SELECTED BY ARCHITECT.

EXHAUST FAN SCHEDULE

MARK	AREA SERVED	BASIS OF DESIGN: MANUFACTURER	BASIS OF DESIGN: MODEL	FAN TYPE	DRIVE	CFM	ESP	RPM	MOTOR HP	VOLT/HZ/PH	WEIGHT	NOTES
EF-01	110 TOILET	GREENHECK	SP-B80	CEILING	DIRECT	75	0.25	900	17 WATTS	115V/60HZ/1Ø	15	1,2,3,4

NOTES:

1. INSTALL IN ACCORDANCE WITH MANUFACTURER WRITTEN INSTRUCTIONS. COORDINATE INSTALLATION WITH ALL TRADES.
2. PROVIDE SPRING HANGER KIT. SUSPEND FAN FROM BOTTOM OF DECK.
3. PROVIDE WITH MANUAL SWITCH AND 15 MINUTE TIMER.

PROJECT NAME / LOCATION:
PROPOSED "INTERIOR AND EXTERIOR ALTERATIONS"
India Pentecostal Assembly
310A SOUTH OYSTER BAY ROAD
SYOSSET, NY 11791
TOWN OF OYSTER BAY

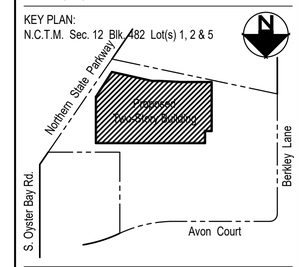
ARCHITECT:
the KENT ARCHITECTURAL COLLABORATIVE, PLLC
132 CLYDE STREET, SUITE 1B
WEST SAYVILLE • NEW YORK • 11796
631.793.9684
rkent.tkac@gmail.com

CONSTRUCTION MANAGER / GENERAL CONTRACTOR:
CORE CONCEPTS CONTRACTING
687 OLD WILLETS PATH, HAUPPAUGE, NY 11788
www.coreconceptscontracting.com 631.435.8985

STRUCTURAL ENGINEER:
reuther&bowen Engineering, Design, Construction Services
326 WARD ST. SCRANTON, PA 18512-2424
PHONE (570) 496-7020 FAX (570) 496-7021

MECHANICAL / ELECTRICAL / PLUMBING ENGINEERS:
DMG designmanagementgroup building systems, smarter
2007 HIGHWAY 315, SUITE 202 PITTSTON, PA 18640
PHONE (570) 299-7520

KEY PLAN:
N.C.T.M. Sec. 12 Blk. 482 Lot(s) 1, 2 & 5



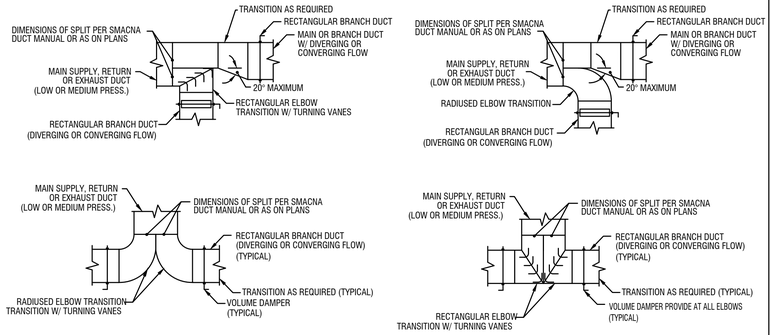
ISSUED FOR CONSTRUCTION CORE & SHELL	2025.03.05
REV PER TOOB COMMENTS	2024.11.22
ISSUED FOR PERMIT	2024.07.31
REVISION:	DATE:

SEAL:

DRAWING TITLE:
MECHANICAL SCHEDULES

DATE: 2024.01.29
DRAWN BY: RK/BMF
PROJECT NO.: 24.03
DRAWING NO.: M7.2

TOB APPLICATION No. 22030355

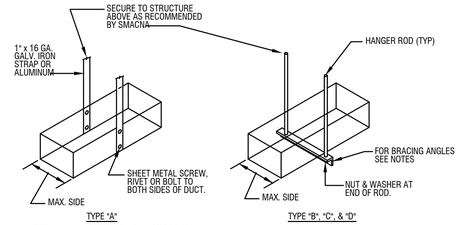


NOTES:

1. FABRICATE SPLIT FITTINGS, ELBOWS, TRANSITIONS, ETC. PER LATEST EDITION OF SMACNA DUCT CONSTRUCTION MANUAL, AS INDICATED ON PLANS, OR AS DESCRIBED IN SPECIFICATION.
2. PROVIDE DUCT SPLIT ARRANGEMENT AS SHOWN ON PLANS.
3. PROVIDE DAMPERS WITH ALL FITTING AS INDICATED ABOVE

1 / M8.1 SPLIT DUCT DETAILS

SCALE: NTS



TYPE "A"
(8 FT. MAX. HANGER SPACING. ALSO PROVIDE 3 HANGERS AT EACH TAKE-OFF OR BRANCH.)

TYPE "B", "C", & "D"
(8 FT. MAX. HANGER SPACING)

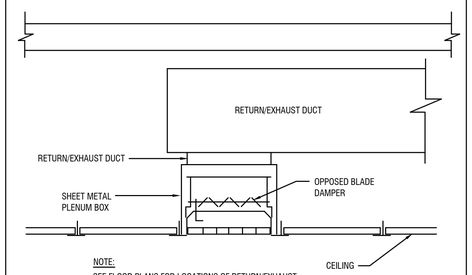
DUCT DIMENSION	HANGER TYPE	ROD DIA.	ANGLE SIZE	MAX SPACING
UP TO 18"	A	1" STRAP	——	8'-0"
19" TO 60"	B	5/16"	1-1/2" x 1-1/2" x 1/8"	8'-0"
61" TO 96"	C	3/8"	1-1/2" X 1-1/2" x 3/16"	8'-0"
OVER 96"	D	1/2"	2" x 2" x 1/4"	4'-0"

NOTES:

1. FOR SEVERAL DUCTS ON ONE HANGER, TYPE "B", "C", OR "D" MAY BE USED. SIZE OF HANGER WILL BE SELECTED ON SUM OF DUCT WIDTHS EQUAL TO MAX. WIDTH OF DUCT SCHEDULE.
2. DO NOT ATTACH DUCT HANGERS TO ROOF DECK OR BOTTOM CORD OF JOISTS. PROVIDE ANGLE BRACING AS REQUIRED.
3. CONNECTIONS TO CONCRETE FLOOR SLABS IS PERMITTED. INSTALL PER SPECIFICATION

2 / M8.1 DUCT HANGERS

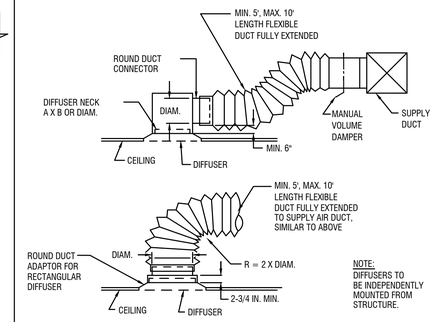
SCALE: NTS



NOTE:
SEE FLOOR PLANS FOR LOCATIONS OF RETURN/EXHAUST AIR REGISTERS & GRILLES.

3 / M8.1 RETURN/EXHAUST AIR GRILLE DUCT DETAIL

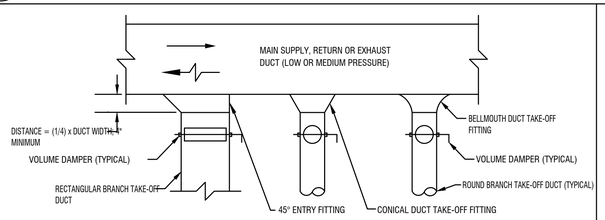
SCALE: NTS



NOTE:
DIFFUSERS TO BE INDEPENDENTLY MOUNTED FROM STRUCTURE.

4 / M8.1 DIFFUSER CONNECTION DETAILS

SCALE: NTS

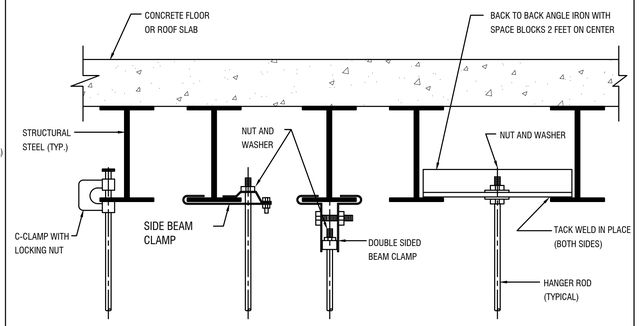


NOTES:

1. SPIN-IN DUCT TAKE-OFF FITTINGS MAY BE USED IN LIEU OF CONICAL OR BELLMOUTH FITTINGS ONLY WHERE MAIN DUCT DIMENSIONS ARE NOT SUFFICIENT TO ALLOW THE USE OF A CONICAL OR BELLMOUTH.
2. SEAL ALL TAKE-OFF AND OTHER DUCT FITTINGS AIR TIGHT AS PER SPECIFICATION.
3. FABRICATE BRANCH DUCT TAKE-OFF FITTINGS PER LATEST EDITION OF SMACNA DUCT CONSTRUCTION MANUAL, AS INDICATED ON PLANS, OR AS DESCRIBED IN THE SPECIFICATION.

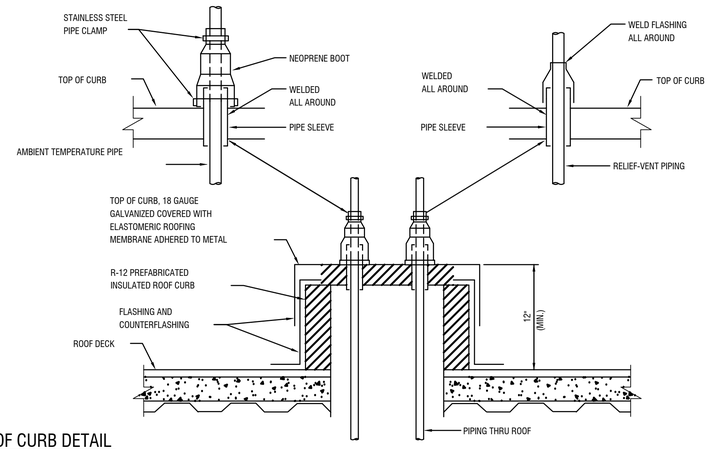
5 / M8.1 BRANCH TAKE-OFF DUCT DETAIL

SCALE: NTS



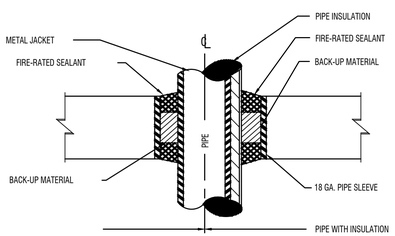
6 / M8.1 METHODS OF HANGER SUPPORT DETAIL

SCALE: NTS



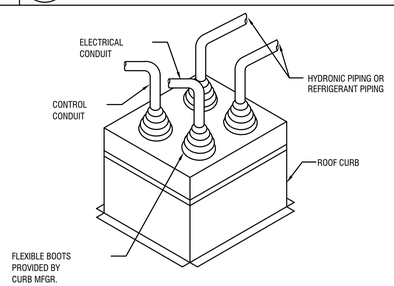
7 / M8.1 PIPING THRU ROOF CURB DETAIL

SCALE: NTS



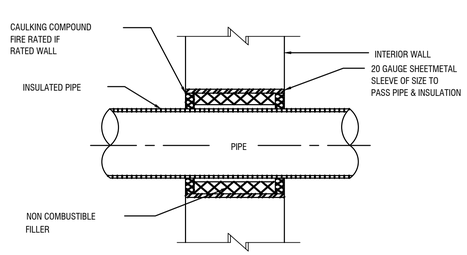
8 / M8.1 PIPE SLEEVE THRU FLOOR DETAIL

SCALE: NTS



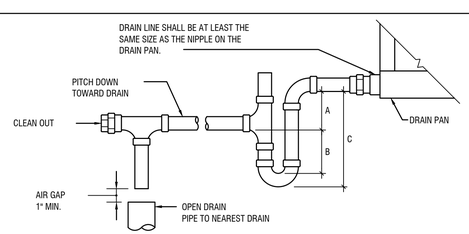
9 / M8.1 PIPING PORTAL THRU ROOF DETAIL

SCALE: NTS



10 / M8.1 PIPE PENETRATION DETAIL (INTERIOR WALL)

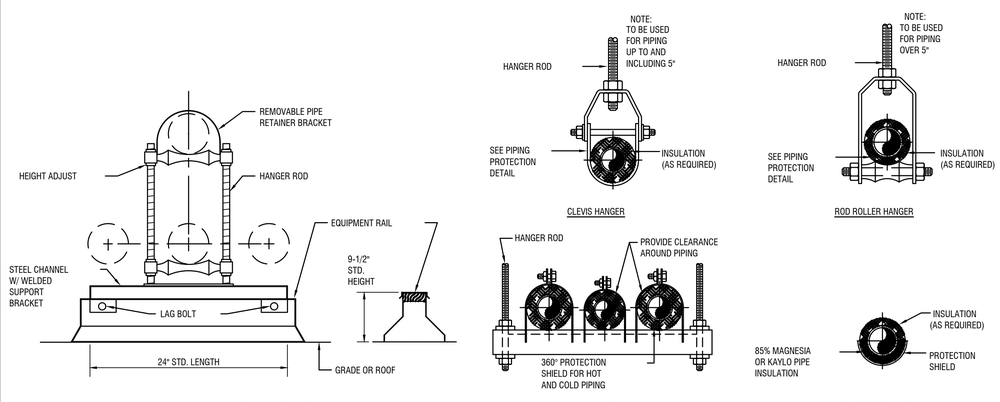
SCALE: NTS



UNIT TYPE	A	B	C
DRAW THRU	(1" FOR EACH 1" OF MAX. NEGATIVE S.P.) + 1"	HALF OF "A"	"A"+"B" + PIPE DIA. + INSULATION
BLOW THRU	1/2" MINIMUM	1/2" PLUS MAX. S.P.	N/A

11 / M8.1 CONDENSATE TRAP DETAIL

SCALE: NTS



12 / M8.1 HVAC PIPE SUPPORT DETAIL

SCALE: NTS

PROJECT NAME / LOCATION:

PROPOSED "INTERIOR AND EXTERIOR ALTERATIONS"
India Pentecostal Assembly
310A SOUTH OYSTER BAY ROAD
SYOSSET, NY 11791
TOWN OF OYSTER BAY

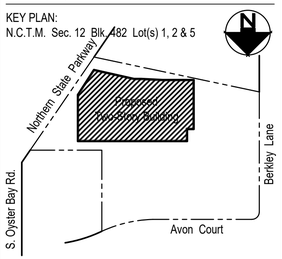
ARCHITECT:
the KENT ARCHITECTURAL COLLABORATIVE, PLLC
132 CLYDE STREET, SUITE 1B
WEST SAYVILLE • NEW YORK • 11796
631.793.9684
rkent.tkac@gmail.com

CONSTRUCTION MANAGER / GENERAL CONTRACTOR:
CCC CORE CONCEPTS CONTRACTING
687 OLD WILLETS PATH, HAUPPAUGE, NY 11788
www.coreconceptscontracting.com 631.435.8985

STRUCTURAL ENGINEER:
reuther&bowen Engineering, Design, Construction Services
326 WARD ST. SCRANTON, PA 18512-2424
PHONE (570) 496-7020 FAX (570) 496-7021

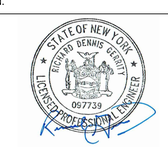
MECHANICAL / ELECTRICAL / PLUMBING ENGINEERS:
DMG designmanagementgroup building systems, smarter
2007 HIGHWAY 315, SUITE 202 PITTSTON, PA 18640
PHONE (570) 299-7520

KEY PLAN:
N.C.T.M. Sec. 12 Blk. 482 Lot(s) 1, 2 & 5



REVISION:	DATE:
ISSUED FOR CONSTRUCTION CORE & SHELL	2025.03.05
REV PER TOOB COMMENTS	2024.11.22
ISSUED FOR PERMIT	2024.07.31

SEAL:



DRAWING TITLE:
MECHANICAL DETAILS

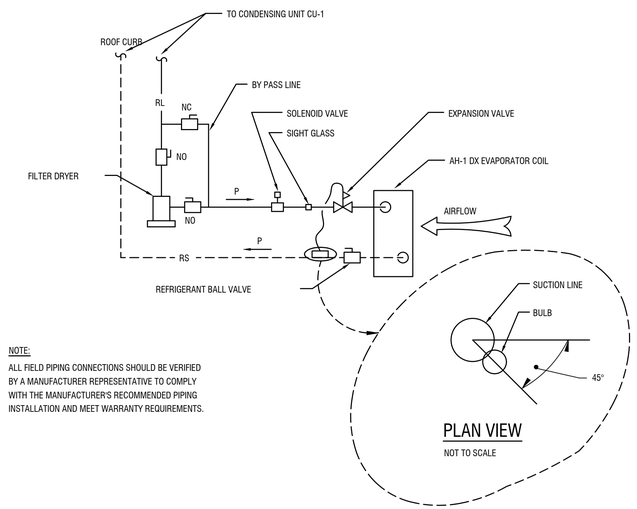
DATE: 2024.01.29

DRAWN BY: RK/BMF

PROJECT NO.: 24.03

DRAWING NO.: M8.1

TOB APPLICATION No. 22030355



NOTE:

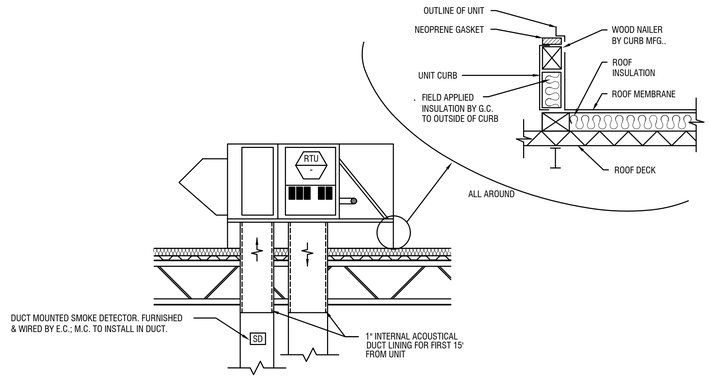
ALL FIELD PIPING CONNECTIONS SHOULD BE VERIFIED BY A MANUFACTURER REPRESENTATIVE TO COMPLY WITH THE MANUFACTURER'S RECOMMENDED PIPING INSTALLATION AND MEET WARRANTY REQUIREMENTS.

PLAN VIEW

NOT TO SCALE

1 M8.2 DX COOLING COIL PIPING DETAIL

SCALE: NTS



ROOFING NOTES (FOR ALL ROOF PENETRATIONS):

1. FINAL APPROVAL TO BE OBTAINED FROM ROOFING MANUFACTURER WHO WILL ULTIMATELY WARRANTY THE INSTALLATION.
2. ALL MATERIALS IN CONTACT WITH THE PIPING INSULATION JACKET SHALL BE SUITABLE FOR 400° F TEMPERATURES
3. FLASHING SHALL BE A MINIMUM OF 6" HIGH WHENEVER POSSIBLE AND MUST BE ABOVE ROOF FLOOD LEVEL.
4. FIELD MEMBRANE AND INSULATION (WHEN APPLICABLE) MUST BE CUT TIGHTLY AROUND THE PROJECTION.

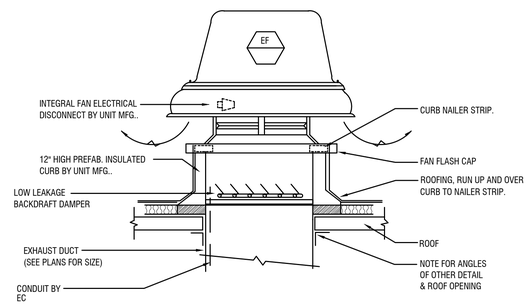
MECHANICAL ROOFTOP UNIT REFURBISHING NOTES:

APPLIES TO EXISTING RTUS:

1. CLEAN CONDENSER COILS
2. CLEAN EVAPORATOR COILS
3. CHANGE ALL FILTERS
4. CHECK AND ADJUST FAN BELT TENSION
5. CHECK COMPRESSOR AMP DRAWS
6. INSPECT REFRIGERATION SYSTEM FOR LEAKS
7. CHECK REFRIGERANT PRESSURE (ADD IF NECESSARY)
8. CLEAN FAN SCROLL AND FAN
9. CLEAN CONDENSATE DRAIN PAN
10. CHECK CONDENSER FANS AND PRESSURE/TEMPERATURE SWITCHES FOR PROPER OPERATION
11. CLEAN AND INSPECT CONDENSATE TRAPS
12. CLEAN AND INSPECT GAS FURNACE FOR PROPER OPERATION
13. CHECK FUEL GAS PRESSURE INTO UNIT.

2 M8.2 ROOFTOP AIR CONDITIONING UNIT

SCALE: NTS

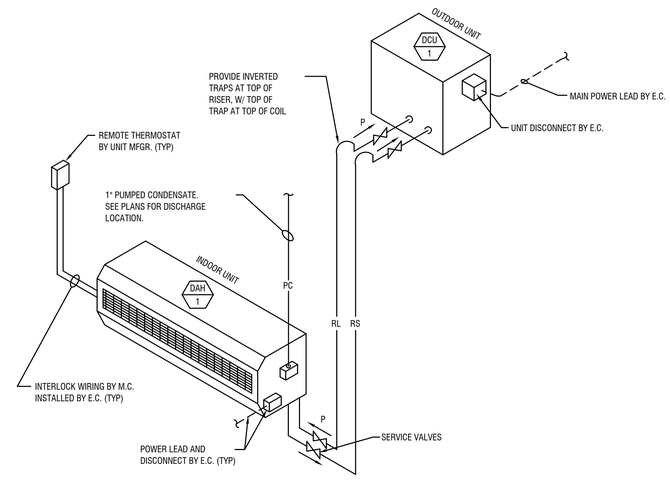


ROOFING NOTES (FOR ALL ROOF PENETRATIONS):

1. FINAL APPROVAL TO BE OBTAINED FROM ROOFING MANUFACTURER WHO WILL ULTIMATELY WARRANTY THE INSTALLATION.
2. ALL MATERIALS IN CONTACT WITH THE PIPING INSULATION JACKET SHALL BE SUITABLE FOR 400° F TEMPERATURES
3. FLASHING SHALL BE A MINIMUM OF 6" HIGH WHENEVER POSSIBLE AND MUST BE ABOVE ROOF FLOOD LEVEL.
4. FIELD MEMBRANE AND INSULATION (WHEN APPLICABLE) MUST BE CUT TIGHTLY AROUND THE PROJECTION.

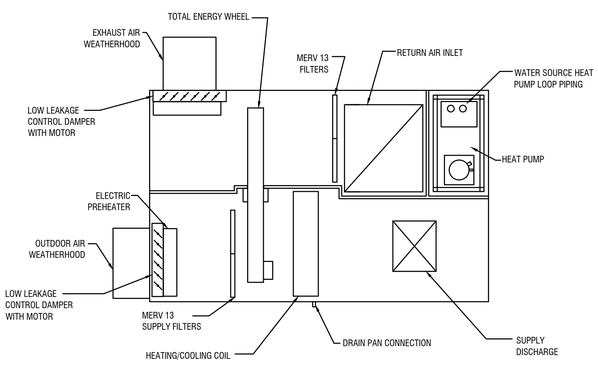
3 M8.2 ROOF EXHAUST FAN DETAIL

SCALE: NONE

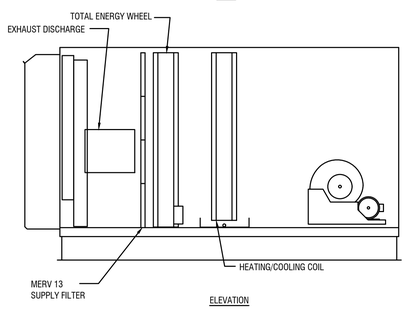


4 M8.2 DUCTLESS SPLIT SYSTEM REFRIGERANT PIPING DETAIL

SCALE: NTS



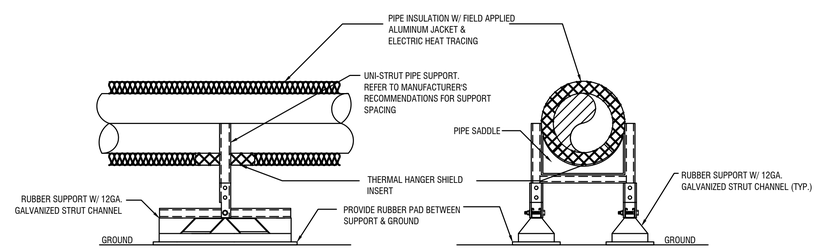
PLAN



ELEVATION

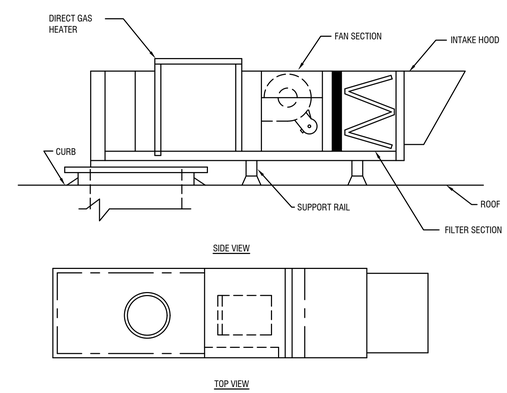
5 M8.2 ENERGY RECOVERY UNIT DETAIL

SCALE: NTS



6 M8.2 PIPE SUPPORT DETAIL

SCALE: NTS



SIDE VIEW

TOP VIEW

7 M8.2 VENTILATION MAKE-UP UNIT

SCALE: NTS

PROJECT NAME / LOCATION:

PROPOSED "INTERIOR AND EXTERIOR ALTERATIONS"

India Pentecostal Assembly

310A SOUTH OYSTER BAY ROAD
SYOSSET, NY 11791

TOWN OF OYSTER BAY

ARCHITECT:

the KENT ARCHITECTURAL COLLABORATIVE, PLLC
132 CLYDE STREET, SUITE 1B
WEST SAYVILLE • NEW YORK • 11796
631.793.9684
rkent.tkac@gmail.com

CONSTRUCTION MANAGER / GENERAL CONTRACTOR:

CCC CORE CONCEPTS CONTRACTING
687 OLD WILLETS PATH, HAUPPAUGE, NY 11788
www.coreconceptscontracting.com 631.435.8985

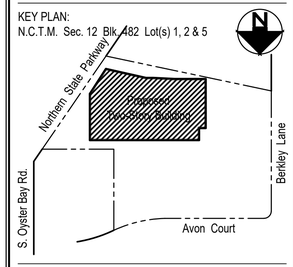
STRUCTURAL ENGINEER:

reuther&bowen Engineering, Design, Construction Services
326 WARD ST. SCRANTON, PA 18512-2424
PHONE (570) 496-7020 FAX (570) 496-7021

MECHANICAL / ELECTRICAL / PLUMBING ENGINEERS:

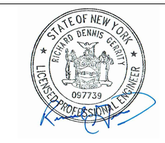
DMG designmanagementgroup building systems, smarter
2007 HIGHWAY 315, SUITE 202 PITTSTON, PA 18640
PHONE (570) 299-7520

KEY PLAN:
N.C.T.M. Sec. 12 Blk. 482 Lot(s) 1, 2 & 5



REVISION:	DATE:
ISSUED FOR CONSTRUCTION CORE & SHELL	2025.03.05
REV PER TOOB COMMENTS	2024.11.22
ISSUED FOR PERMIT	2024.07.31

SEAL:



DRAWING TITLE:

MECHANICAL DETAILS

DATE: 2024.01.29

DRAWN BY: RK/BMF

PROJECT NO.: 24.03

DRAWING NO.: M8.2

TOB APPLICATION No. 22030355